



# **Ø Particles and Matter**

*The mass of matter, the location of the antimatter,  
and the surplus that is everything*



For the reader who chose to read.

Thank you.



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# Artist's Note

This book, *Particles and Matter*, is the fifth notebook in the Ø Artist's Proofs catalogue of The 420 Code. Where Notebook IV (Forces and Constants) derived the gauge structure of the Standard Model and the values of the fundamental constants, Notebook V asks what those constants weigh. It is the first time the 420 Code touches ordinary matter directly: a proton mass, the missing antimatter, and the surplus that survived. All three are read from the same four axioms {S, B, R, C} and the single axiom  $1:1 + 1 \times \epsilon @ AS$ .

The book has four chapters, and they form one argument read four ways.

Chapter 1 (The Resistance, AP30) weighs the matter. Mass is reframed as geometric resistance — how many ways the break has to hold itself against the substrate — and the proton-to-electron mass ratio is reconstructed from the structural integers {21, 3, 4} and one measured input,  $\alpha$ . The integer decomposition  $1764 + 63 + 9 = 1836$  lands within five parts per billion of measurement at order  $\alpha$ , with zero free parameters.

Chapter 2 (The Flip, AP47) prices the neutron honestly, after the first prediction died. On 2 August 2026 the bare neutron-proton row of Chapter 1 fired at  $7.24\sigma$ , and the corpse stays on

the board. The Flip finds what that row had priced as free: a neutron holds a distinction — one of its three couplings sits in the other orientation — and holding costs, at second order, one held grain leaking at the substrate's rate through the pair's eight dimensional expressions. The realised difference,  $\Delta_{\text{real}} = \Delta_{\text{bare}} / (1 + \alpha^2 / 8\pi) = 2.53098857035 m_e$ , lands at  $-0.005\sigma$  and is frozen beside the corpse, never in its place.

Chapter 3 (The Ledger, AP22) locates the antimatter.

Baryogenesis is read not as a process in the early universe but as topology: the event horizon is the  $\sigma$ -boundary of Axiom S, the involution  $\sigma$  is the Tomita–Takesaki modular conjugation, and the antimatter is segregated behind every horizon. The break  $\varepsilon$  has no  $\sigma$ -image — which is why the ledger never closes, and why there is something rather than nothing.

Chapter 4 (The Surplus, AP26) reads the leftover. The axiom, read as an energy partition, gives the baryon asymmetry its form:  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ , the ash of the break. The form is derived; the numerical value of  $E(\varepsilon)$  is the sharpest open debt in the body of work.

The four meet on one object. The break  $\varepsilon$  is read as resistance (the mass, Chapter 1), as the cost of a held distinction (the neutron, Chapter 2), as the unmirrored entry (where the antimatter went, Chapter 3), and as energy (the surplus, Chapter 4). These are not four results but one break under

four projections — the manifold-scale analogue of the bridge equation that united Notebook IV.

Notebook V engages nineteen kill switches — the formal falsification handles for every load-bearing claim — recorded in the Master Kill Switch Registry (v5.27, 4 September 2026). One has fired: KS-NPP.1, the bare neutron-proton row of Chapter 1, on 2 August 2026 at  $7.24\sigma$  — shown, never repaired, counted outside the live total; AP47's KS-FLIP.1 is frozen beside it (Chapter 2), with KS-30.2 cross-listed there and counted once. None is closed within these four chapters: KS-46A is derived, and KS-46B and KS-46C are addressed (Chapter 3 — KS-46B's bifurcation tension is discharged via Sewell 1982, but the full AQFT bridge retains gaps, so the switch is addressed, not closed); the rest are live, structural or empirical, and named openly.

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# Orientation

## Where Notebook V fits

The 420 Code is organised across eight bound notebooks. Each notebook is a standalone book carrying one division of the architecture; the eight together carry the full body of work. Notebook V depends on Notebook I for the axioms and the {S, B, R, C} conditions; on Notebook II for spacetime and the manifold (AP20); on Notebook III for the quantum record; and on Notebook IV for the forces and constants — in particular AP06 (the leakage constant  $\alpha$ ), AP24 (the six-face reading of  $\epsilon$ ), AP28 (the channel count and the value of G), and AP25 (the Born rule). It opens the matter sector that those notebooks make possible.

The reading order within Notebook V is front to back. Chapter 1 weighs the matter and sets the mass scale. Chapter 2 prices the neutron's held distinction beside the row that fired. Chapter 3 segregates the antimatter and proves the break has no mirror. Chapter 4 reads the surplus that the segregation protects and the mass scale sizes. Chapter 3 is depended on by Chapter 4 for the topological protection of the surplus; Chapter 2 stands on Chapter 1; Chapter 1 is independent of the rest but opens the sector.

## How to read this book

The book alternates two voices as the material requires. Narrative passages carry the structural reading — what the axiom forces and why — in plain language; formal passages carry the lemmas, propositions, and the arithmetic a physicist can check. Each chapter keeps its own architecture: an Artist's Note opens it, a Dependency-and-Scope section states what it does and what it owes, the numbered body carries the derivation, and a Kill Switches section names the falsification handles before the chapter closes.

## A note on notation across chapters

Three notational items recur and are worth flagging at the door. The symbol  $\varepsilon$  has a single meaning across all four chapters: the unique unpaired element of the axiom  $1:1 + 1 \times \varepsilon$  @ AS — the break. But  $\varepsilon$  has several faces, and the faces are distinct numbers belonging to one object (AP24): its coupling face is the leakage constant  $\varepsilon_{\text{leak}} = \alpha \approx 1/137$  (AP06, read in Chapter 1), and its energy face is  $E(\varepsilon)$ , the dimensionless energy of the break that sets the baryon asymmetry (Chapter 3). These two numbers differ by orders of magnitude; why one break carries faces so far apart is itself part of what AP24's self-consistency must explain (KS-60). The symbol  $\sigma$  is the orientation-reversal involution of Axiom S, identified at the horizon with the Tomita–Takesaki modular conjugation

(Chapter 2). The annotation @ AS marks the axiom as read on the Actualization State — the manifold of AP20.







## Chapter 1

# The Resistance

*The proton mass derived to five parts per billion*

*one measured input · zero free parameters*

*Artist's Proof 30*

# Artist's Note

*What this paper is, and why.*

You have weight. Right now, pressing down into the chair. That weight is not a substance the universe handed you. It is resistance — the structural cost of holding a record together against a substrate that would rather let it close.

This paper takes the heaviest ordinary thing inside your body, the proton, and derives its mass from the structure of the break. Not measures it — derives it. The proton is 1836 times heavier than the electron, a number measured to extraordinary precision and, until now, never derived. One measured input,  $\alpha \approx 1/137$ . Zero free parameters. The proton-to-electron mass ratio falls out of the manifold's geometry to five parts per billion.

Mass, here, is geometric resistance — how many ways the break has to hold, and how hard it has to work to keep holding. The electron is the simplest charged break: one channel, the unit of mass. The proton integrates all twenty-one. It has more knots. That is what mass is.

AP30 stands on the constants sector of Notebook IV: the leakage constant (AP06), the residual that reads every constant as a face of one break (AP24), and the channel count that derived gravity (AP28). It is the first mass ratio the body

of work derives from {S, B, R, C}. If the method holds, the rest of the mass spectrum is in reach. If it fails, the kill switches say where.

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# Orientation

## Where this paper fits in the 420 Code

AP30 is the first chapter of Notebook V — Particles and Matter. The break has produced spacetime (Notebook II), the quantum record (Notebook III), and the forces and constants (Notebook IV). Notebook V is where those constants make the thing they weigh: matter. AP30 derives the proton's mass. AP22 (The Ledger) accounts for the antimatter. AP26 (The Surplus) reads the leftover. Read in that order.

AP30 reads after Notebook IV. It inherits the channel count from AP28 (The Constant), the six-face residual from AP24 (The Residual), and the leakage constant  $\alpha$  from AP06 (The Leakage Constant).

## How to read this paper

Two voices alternate as the material requires. The lemmas and the proposition speak in the precision the mathematics requires — the channel counts, the perturbative series, the numbers — preserved exactly. The connective passages speak to you directly. The gear-changes are intentional. The integer comes first: 1836, the static geometry. The fractional corrections follow — the dynamic maintenance, its leakage,

and the residual beneath them — each order smaller than the last by roughly  $\alpha$ .

## 0 — Dependency and Scope

### 0.1 What this paper does

AP30 derives the proton-to-electron mass ratio,  $m_p/m_e$ , from the four axioms {S, B, R, C} and one measured input,  $\alpha \approx 1/137$ . It reframes mass as geometric resistance, decomposes the integer 1836 into three independent static layers, adds the dynamic corrections the break's self-maintenance forces, and lands within five parts per billion of the measured value at order  $\alpha$ , with zero free parameters. The  $O(\alpha^2)$  term reduces the residual to about 0.008 parts per billion; the coefficient at that order (16) is carried in the verified formula but its structural decomposition from {21, 3, 4} is not yet established and is owed under KS-30.3, so the parameter-free claim is stated firmly at order  $\alpha$ .

### 0.2 Dependencies

AP06 (The Leakage Constant):  $\alpha$  as the leakage rate and the perturbative expansion parameter. AP10 (The Dimension): the structural integers 21, 3, 4. AP19 (The Direction): the SU(3) three-face geometry behind Layers 2 and 3. AP20 (The Proof): axiom independence, which Lemma 1 rests on. AP24 (The Residual): the six-face reading of the constants. AP28 (The Constant): the 21-channel count and the  $\alpha$ -G connection.



AP29 (The Actualization Proof): mass as  $\kappa(B) \times AS$ . AP03 (The Ratio): the three fundamental quantities  $c$ ,  $\hbar$ ,  $G$ .

### 0.3 Axiom mapping

The three static resistance layers trace to distinct axiom subsets. Layer 1 (manifold capacity) is how the break persists geometrically across the channels and dimensions —  $B$  read against the manifold the four axioms resolve. Layer 2 (face projection) is what the break persists  $AS$ , projected onto the three-face strong-force geometry — the  $S$ -derived faces. Layer 3 (exchange matrix) is how the bound state stays bounded — the continuous mutual exchange Axiom  $S$  requires. The dynamic terms are the cost of sustaining the break against the substrate's leakage ( $\alpha$ , AP06) across the propagation geometry ( $C$ ).

### 0.4 Epistemic status per section

§1 Definitions: framing and definition. §2 Lemma 1: structural argument (additivity from axiom independence; KS-30.1 live). §3 Lemma 2: derivation (the integer count  $1764 + 63 + 9 = 1836$ ). §4 Lemma 3: derivation (the dynamic maintenance term). §5 Lemma 4: derivation (the isotropic-leakage correction). §6 Proposition: derivation (the full ratio and its precision). §7 The Structural Residual: synthesis — the residual read as  $\varepsilon$  itself. §8 Connections: the 420 Code integration.

## 0.5 Kill switch summary

Seven kill switches engage on this paper, all live. KS-30.1 (additivity formalisation), KS-30.2 (leakage isotropy), KS-30.3 (higher-order terms), and KS-30.4 (alternative formula) test the proton-mass derivation directly. KS-NPP.1, KS-NPP.2, and KS-NPP.3 test the neutron-proton mass difference (decomposition, sign, lattice-QCD consistency). KS-30.4 and the three KS-NPP switches were registered against AP30 through the Ø Predictions integration; their operational tests are carried in Ø Predictions (Part I and Part III) per the Master Kill Switch Registry.

## 0.6 Structural debts owed

The additivity of the resistance layers is argued from axiom independence but not yet exhibited as a Cartesian-product theorem (tracked as KS-30.1). The extension to the rest of the mass spectrum — muon, tau, the quark hierarchy, the three-generation structure — is owed (it meets the generation debt carried as KS-66 in AP27). The absolute electron-mass energy scale is owed; with  $m_p/m_e$  derived, it is fixed up to one overall scale tied to AP28's G derivation.

## **0.7 Items held open without claiming debt status**

The closed-form expressions for the  $O(\alpha^2)$  and higher orders are not computed; the series is claimed to converge from the perturbative hierarchy (KS-30.3). The absolute energy scale of the electron mass is held open pending the connection to the Planck mass through AP28.

## **0.8 Structural relationships**

AP24 (The Residual), locked in Notebook IV, treats the proton mass as one of six measured faces of  $\varepsilon$ ; AP30 derives that face from {S, B, R, C}, retiring its empirical status. The relationship is framework-and-value — AP30 inherits AP24's six-face reading and supplies the value of one face — not a dependency cycle. AP30 opens the matter sector that AP22 (The Ledger) and AP26 (The Surplus) complete. With AP30 in place, the body of work expresses all fundamental scales —  $c$ ,  $\hbar$ , and  $G$  (AP03, AP28) and now  $m_p/m_e$  — in terms of the axioms and the single measured input.

# 1 — Definitions

## Mass as resistance

Mass is not substance. Mass is geometric resistance — the structural opposition a record offers to displacement, change, or acceleration. A particle with mass  $m$  resists  $m$  times more than a particle with unit mass.

The unit of resistance is set by the minimal break: the electron, mass  $m_e$  (Axiom B, AP06).

## The 420 Code numbers

**21:** the number of geometric channels through which the break persists (AP10 + AP28). These arise from the combinatorial structure of three spatial dimensions resolved from one manifold.

**3:** the number of faces of the manifold (AP10). Identified with the three colours of SU(3) (AP19). Each face corresponds to one of three structural roles: C (Constraint — locality, read as propagation), S (Symmetry, read as exchange), B (Break).

**4:** the number of spacetime dimensions into which the manifold resolves (AP10, AP20). Three spatial + one temporal.

**$\alpha_{em} = 1/137.036$ :** the fine structure constant, identified as  $\varepsilon$  — the leakage constant, the substrate's fundamental coupling

strength (AP06). This is the rate at which the substrate's minimal break couples to its own field.

## 2 — Lemma 1: Additivity of Independent Resistance

**Claim:** When independent structural constraints act on the same bound state, their contributions to the total geometric resistance are additive.

**Proof:** AP20 establishes that the four axioms {S, B, R, C} are independent — no axiom is derivable from the others, and each is categorically forced (no alternative formalisation exists).

The three static resistance layers trace to different axiom subsets:

Each layer answers a structurally distinct question about the bound state. Layer 1 asks how the break persists geometrically (the manifold's capacity).

Layer 2 asks what the break persists AS (its projection onto the strong-force geometry). Layer 3 asks how the bound state remains bounded (its internal exchange).

These questions are independent because they trace to independent axiom subsets acting on independent degrees of freedom.

Independent constraints acting on independent degrees of freedom contribute additively to total resistance.

This follows from the same principle that governs independent energy modes in statistical mechanics: when a system's total state space is the Cartesian product of independent subspaces, the total energy (or resistance) is the sum of contributions from each subspace.

The axiom independence proven in AP20 guarantees the subspaces are independent.

**Therefore the total static resistance is the sum:  $1764 + 63 + 9 = 1836$ .  $\square$  5**

KS-30.1: The Cartesian product structure of the state space is argued from axiom independence, not derived as a theorem. The argument is strong but not fully formalised.

### **3 — Lemma 2: The Static Resistance Count**

#### **Layer 1 — Manifold capacity: $21^2 \times 4 = 1764$**

The manifold has 21 geometric channels (AP10, AP28) and resolves into 4 spacetime dimensions (AP10, AP20). Each channel must express through each dimension, giving  $21 \times 4 = 84$  dimensional expressions.

The proton integrates all 21 channels. It is not a 1-channel object like the electron.

Each of the 84 dimensional expressions couples to each of the 21 channels, generating the full resistance network:  $84 \times 21 = 21^2 \times 4 = 1764$ .

This is the dominant term. It says: the proton resists 1764 times more than the bare break because it integrates the full channel network across all spacetime dimensions.

#### **Layer 2 — Face projection: $21 \times 3 = 63$**

The proton forms under SU(3) geometry (AP19): three faces corresponding to three colours. The 21 channels must anchor to these three specific faces.



This is the resistance cost of projecting the full manifold onto the 3-colour strong-force substrate. Each channel anchored to each face:  $21 \times 3 = 63$ .

This layer is a constraint ON the manifold capacity (Layer 1). The channels are already counted. Layer 2 adds the resistance of locking them to three specific faces rather than leaving them unanchored.

### **Layer 3 — Exchange matrix: $3^2 = 9$**

By Axiom S, the three faces must maintain continuous mutual exchange to remain bounded. The full interaction matrix of three faces is  $3 \times 3 = 9$  pathways.

This maps exactly to  $U(3)$ : 8 gluon generators + 1 colour singlet required for total confinement.

This is the smallest static term. It counts only the face-to-face exchange overhead, not the channel structure within each face.

### **Uniqueness of the construction**

All three-term sums of structural products (products of powers of  $\{21, 3, 4\}$ ) that equal 1836 were enumerated:

Among them, exactly one is hierarchically nested — each successive term using fewer body of work numbers and corresponding to a distinct geometric feature (manifold

capacity, face projection, exchange matrix). This hierarchical decomposition is the one used here; whether it is the unique admissible decomposition under a precise enumeration constraint is exactly what KS-30.4 holds open, and the constraint set and the count of admissible decompositions under it are not exhibited in this paper.

The construction is not arbitrary. It is the hierarchical decomposition this paper adopts; its uniqueness among hierarchically nested decompositions is asserted, not exhibited (KS-30.4).

## 4 — Lemma 3: Dynamic Maintenance

**Claim:** the bound state continuously sustains itself against substrate leakage at rate  $\alpha_{em}$  across all 21 channels, contributing  $\alpha_{em} \times 21$  to the mass ratio.

The integer 1836 is the static geometry — the resistance of the structure at rest. But the proton is not static internally. Three quarks must continuously exchange gluons, maintain confinement, and sustain the bound state.

This requires ongoing coupling to the substrate.

**Why  $\alpha_{em}$ , not  $\alpha_s$ ?** The strong coupling  $\alpha_s$  describes how quarks interact WITH EACH OTHER inside the proton. The dynamic maintenance cost is something different: it is the proton's coupling to the SUBSTRATE.

The proton sustains itself against the substrate's leakage — against the tendency of the break to heal. The substrate's leakage rate is  $\alpha_{em}$  (AP06: the leakage constant).

All other coupling constants ( $\alpha_s$ ,  $\alpha_G$ ) are derived from  $\alpha_{em}$  through the channel geometry. The proton does not maintain itself against the strong force — it IS a strong-force object.

It maintains itself against the substrate trying to close the break. The substrate speaks at  $\alpha_{em}$ .

The coupling must span all 21 channels because the proton integrates all 21 (Layer 1). Each channel contributes  $\alpha_{em}$  to the total dynamic coupling. The dynamic maintenance term is therefore  $\alpha_{em} \times 21 = 0.15324$ .

## 5 — Lemma 4: Isotropic Leakage

**Claim:** the dynamic coupling does not remain perfectly within the channels. It leaks isotropically into the 3D substrate, reducing the effective dynamic term by a factor of  $1/(84\pi)$ .

The 21-channel dynamic coupling radiates into three-dimensional space (AP10: three spatial dimensions). In 3D, a source radiating isotropically distributes energy across a solid angle of  $4\pi$  steradians.

The dynamic coupling has 84 dimensional expressions (21 channels  $\times$  4 dimensions, from Layer 1). Each dimensional expression leaks into the full solid angle.

The leakage per dimensional expression per solid angle is  $1/(84\pi)$ . The total leakage fraction of the dynamic coupling is:

$$\text{Leakage fraction} = 1/(84\pi) = 0.00379$$

The effective dynamic term is:

$$\alpha_{\text{em}} \times 21 \times (1 - 1/(84\pi)) = 0.15266$$

The 84 that appears in the leakage denominator is the same 84 from the static count (Layer 1: 21 channels  $\times$  4 dimensions).

The manifold's dimensional structure both generates the static resistance and normalises the leakage. This is not a

coincidence. The dimensional expressions are the channels through which the leakage occurs.

The leakage is normalised by the very structure it leaks through.

KS-30.2: The assumption of perfectly isotropic leakage gives  $1/(84\pi)$  exactly. Anisotropic leakage (preferential radiation along specific channels or faces) would modify the correction.

The 5 ppb residual is consistent with a small anisotropic contribution at  $O(\alpha^2_{em})$ . See §8.

## 6 — Proposition: The Proton–Electron Mass Ratio

### From Lemmas 1–4:

$$m_p / m_e = 1836 + \alpha_{em} \times 21 \times (1 - 1/(84\pi)) + \alpha^2_{em} \times 21 \times 16/1836 + O(\alpha^3_{em})$$

**Static resistance** (Lemmas 1–2): three independent layers from three axiom subsets, summing to 1836.

**Dynamic maintenance** (Lemma 3): substrate coupling across all channels at rate  $\alpha_{em}$ , giving  $\alpha_{em} \times 21$ .

**Isotropic leakage** (Lemma 4): radiation loss normalised by the dimensional expression count and the solid angle, reducing the dynamic term by factor  $(1 - 1/(84\pi))$ .

**$O(\alpha^2_{em})$  correction:** the leakage of the leakage — the break operating on its own maintenance.

The coefficient  $21 \times 16/1836$  arises from the same channel geometry (21) acting on the ratio of a dimension-count factor (16, whose structural decomposition from {21, 3, 4} is owed under KS-30.3 — it is not Layer 3, which is 9) to the static resistance (1836).

This term is structurally required by the recursive nature of 1:1 +  $1 \times \varepsilon$  @ AS: the break breaks its own repair.

**$O(\alpha^3_{\text{em}})$  and higher:** each successive order is smaller by a factor of roughly  $\alpha_{\text{em}}$ . The series converges but never terminates. This is  $\varepsilon$  operating recursively.

Numerical evaluation:

$$1836 + (0.0072974 \times 21 \times 0.99621) + (0.0072974^2 \times 21 \times 16/1836)$$

$$= 1836 + 0.15266370 + 0.00000975 = 1836.15267344$$

Measured (CODATA 2022): 1836.15267343(11)

Residual:  $+0.000000014 \approx 0.008$  parts per billion — 7.5× smaller than the experimental uncertainty ( $\pm 0.06$  ppb).

*(The factors in the line above are shown rounded to the displayed digits; the products and the final ratio are evaluated at full precision —  $\alpha = 0.00729735$ ,  $1 - 1/(84\pi) = 0.99621060$ .)*

## Precision

The  $O(1)$  term captures 99.992% of the ratio. The  $O(\alpha_{\text{em}})$  correction captures the remaining 0.008%. The  $O(\alpha^2_{\text{em}})$  correction captures the residual to within 0.008 ppb — below the current experimental uncertainty — using a coefficient whose structural decomposition is owed under KS-30.3; the parameter-free claim is therefore stated firmly at order  $\alpha$ .



Each successive order in the perturbative expansion is smaller by a factor of roughly  $\alpha_{\text{em}}$ . This is the signature of a natural perturbative series, not a fitted approximation.

## 7 — The Structural Residual

The measured proton-electron mass ratio exceeds the  $O(\alpha_{em})$  prediction by approximately 0.000001. This residual is of order  $\alpha^2_{em}$  — the next term in the perturbative expansion.

**The governing axiom 1:1 +  $1 \times \epsilon$  @ AS demands this residual exist.**

Perfect symmetry plus minimal break. Nothing closes perfectly. The formula expresses the proton's geometric resistance in terms of the manifold's structural constants with extraordinary precision. But the proton IS a break.

Its existence means the substrate has not closed. The formula describes the structure of the break, but it cannot describe the break completely, because the break is what makes the description possible.

Each order in the perturbative expansion captures more of the structure:

**$O(1)$ :** the static geometry — the integer resistance count (1836).

**$O(\alpha_{em})$ :** the dynamic breathing and its leakage — the cost of sustaining the geometry (0.15266).

**$O(\alpha^2_{em})$ :** the leakage of the leakage — the break operating on its own maintenance ( $\sim 0.000001$ ).

**$O(\alpha^3_{em}), O(\alpha^4_{em}), \dots$** : each order catches a finer correction. The series converges but never terminates.

This is  $1:1 + 1 \times \varepsilon$  @ AS operating recursively: the break breaks its own repair, which breaks the break of the repair, ad infinitum. Each layer adds roughly one factor of  $\alpha_{em}$  ( $\approx 1/137$ ) in precision.

The residual is not noise. It is not experimental error. It is not a missing term.

It is  $\varepsilon$  itself — the minimal break, persisting at the bottom of every derivation, exactly as the axiom demands.

The proof that the formula works IS the proof that it can never be perfect.  $1:1 + 1 \times \varepsilon$  @ AS. Always.

## 8 — Connections

### AP06 (The Leakage Constant)

$\alpha_{em}$  as the leakage constant provides both the dynamic maintenance rate and the perturbative expansion parameter. The proton's mass is fundamentally set by the substrate's leakage rate operating across the channel geometry.

### AP10 (The Dimension)

The numbers 21, 3, and 4 all originate in AP10. Three spatial dimensions  $\rightarrow$  three faces  $\rightarrow$  21 channels. Four spacetime dimensions. The entire formula is downstream of AP10's geometry.

### AP19 (The Direction)

SU(3) from three faces. The face projection (Layer 2) and the exchange matrix (Layer 3) both arise from AP19's derivation of the strong interaction from the manifold's three-face structure.

### AP20 (The Proof)

Axiom independence. The additivity of resistance layers (Lemma 1) depends on the independence of {S, B, R, C} proven in AP20.

## AP28 (The Constant)

The 21-channel count. The G derivation. The connection between  $\alpha_{em}$  and the gravitational constant. AP30 and AP28 share the same structural integers because they describe the same manifold from different angles.

## AP29 (The Actualization Proof)

Coupling capacity  $\kappa(B)$ . AP29 establishes that mass is  $\kappa(B) \times AS$  — coupling capacity actualised. AP30 gives the specific resistance count for the proton:  $\kappa(B)$  for the 3-face, 21-channel bound state equals  $1836 + O(\alpha_{em})$ .

## The Fusion Programme

The binding energy ceiling  $BE/A \approx \alpha_{em} \times (1+1/\pi) \times m_N$  (Clarification 6) no longer contains an empirical mass. With  $m_N/m_e$  derived, the ceiling is fully expressed in terms of  $\{S, B, R, C\}$ :

$$BE/A_{ceiling} = \alpha_{em} \times (1+1/\pi) \times [1836 + \alpha_{em} \times 21 \times (1 - 1/(84\pi))] \times m_e$$

No free parameters remain in the nuclear binding energy ceiling.

## The three fundamental quantities

AP03 establishes three fundamental quantities:  $c$  (propagation),  $\hbar$  (record),  $G$  (persistence). AP28 derives  $G$  from  $\alpha_{em}$ ,  $m_e$ ,  $\hbar$ , and  $c$ . AP30 derives  $m_p$  from  $\alpha_{em}$ ,  $m_e$ , and manifold geometry.

Together with AP03, the axioms now express all fundamental scales in terms of the axioms.

# Kill Switches

Kill switch numbers are globally unique across the body of work. Seven switches engage on this paper; all are live.

## KS-30.1 — Additivity formalisation

**Claim.** Independent structural constraints acting on independent degrees of freedom contribute additively to total geometric resistance, giving the static sum  $1764 + 63 + 9 = 1836$ .

**Test.** Show that the axioms require a non-additive combination rule for the three resistance layers, or that the layers do not span independent subspaces.

**Status.** LIVE — STRUCTURAL.

**Recovery.** The integer 1836 and the additive decomposition fail; the static count must be rebuilt from the correct combination rule. The Cartesian-product structure is argued from axiom independence (AP20), not yet exhibited as a theorem — the standing debt of this paper.

## KS-30.2 — Leakage isotropy

**Claim.** The dynamic coupling leaks isotropically into the 3D substrate, normalised by the solid angle, giving the factor  $1/(84\pi)$  exactly.

**Test.** Show the leakage is anisotropic — preferential radiation along specific channels or faces.

**Status.** LIVE — EMPIRICAL.

**Recovery.** The  $1/(84\pi)$  correction changes; the  $O(\alpha)$  term is modified by the anisotropy. The residual at  $O(\alpha^2)$  is consistent with a small anisotropic contribution but this has not been derived.

## KS-30.3 — Higher-order terms

**Claim.** The perturbative series converges, each successive order smaller by roughly  $\alpha$ ; the residual is  $\varepsilon$  operating recursively.

**Test.** Compute the explicit  $O(\alpha^2)$ ,  $O(\alpha^3)$  forms and show they misbehave — do not shrink, or do not close.

**Status.** LIVE — STRUCTURAL.

**Recovery.** The convergence claim is limited to the orders explicitly computed; a closed form for all orders is not yet derived.



## KS-30.4 — Alternative formula

**Claim.** The hierarchical decomposition of 1836 from {21, 3, 4} is the unique structural construction.

**Test.** Exhibit a different formula in the 420 Code numbers that matches  $m_p/m_e$  to equal precision with a different structural interpretation.

**Status.** LIVE — STRUCTURAL. Registered against AP30 via Ø Predictions (Part I); operational test carried there per the Master Kill Switch Registry.

**Recovery.** The derivation is underdetermined; the structural reading becomes one of several candidates.

## KS-NPP.1 — Two-component decomposition (neutron-proton mass)

**Claim.** The neutron-proton mass difference decomposes as  $(m_n - m_p)/m_e = 3(1 - 1/(2\pi)) + \alpha(1 + 1/(2\pi)) = 2.53099$ , matching the measured value to 2 ppm.

**Status at the wave, 4 September 2026.** KS-NPP.1 FIRED — 2 August 2026. Against CODATA 2022,  $2.530988574(74) m_e$ , the bare row  $2.53099393 m_e$  sits 2.12 ppm high on a 0.288 ppm bar —  $7.24\sigma$ . No repair: the required correction is negative, every term of the formula is positive. The corpse stays on the board. AP47 (The Flip, Chapter 2) prices what this

row had priced as free — a held distinction at second order — and freezes  $\Delta_{\text{real}} = \Delta_{\text{bare}}/(1 + \alpha^2/8\pi) = 2.53098857035 \text{ m}_e$  ( $-0.005\sigma$ ) beside it as KS-FLIP.1, never in its place. KS-NPP.2 and KS-NPP.3 are unaffected.

**Test.** Show the two-component decomposition is not forced by the structure.

**Status.** LIVE — STRUCTURAL. Registered against AP30 via  $\emptyset$  Predictions (Part III) per the Master Kill Switch Registry.

**Recovery.** The neutron-proton form fails; the mass-difference must be rebuilt.

## KS-NPP.2 — Sign of the difference

**Claim.** The structure forces the neutron heavier than the proton (the sign of the mass difference).

**Test.** Show the sign is not determined by the geometry.

**Status.** LIVE — STRUCTURAL. Registered against AP30 via  $\emptyset$  Predictions (Part III) per the Master Kill Switch Registry.

**Recovery.** The structural account of the sign fails.

## KS-NPP.3 — Lattice QCD consistency

**Claim.** The derived neutron-proton difference ( $2.53099 \text{ m}_e$ , 2 ppm) is consistent with lattice QCD.

**Dated note, 4 September 2026.** The value quoted here is the bare row, fired 2 August 2026 (see KS-NPP.1 above); the realised row of AP47, 2.53098857035 m<sub>e</sub>, is the one to compare with lattice QCD at the re-read.

**Test.** A lattice-QCD result inconsistent with the derived value.

**Status.** LIVE — STRUCTURAL. Registered against AP30 via Ø Predictions (Part III) per the Master Kill Switch Registry.

**Recovery.** The consistency claim fails; the derivation conflicts with first-principles QCD.

# **What This Establishes and What Remains Open**

## **What this paper closes**

The proton-to-electron mass ratio is derived from  $\{S, B, R, C\}$  and one measured input, to within five parts per billion at order  $\alpha$  with zero free parameters; the  $O(\alpha^2)$  term reduces the residual to about 0.008 parts per billion, using a coefficient whose structural decomposition is owed under KS-30.3. Mass is established as geometric resistance, and the integer 1836 is decomposed into three independent static layers, each tracing to a distinct axiom subset.

## **What this paper does not close**

The absolute electron-mass energy scale is not fixed by this paper. The rest of the mass spectrum — muon, tau, the quark hierarchy, the three-generation structure — is not derived; the method is conjectured to extend. The additivity of the resistance layers is argued, not formally proved (KS-30.1).

## **Items held open without claiming debt status**

The closed-form expressions for the  $O(\alpha^2)$  and higher orders (KS-30.3). The absolute energy scale of the electron mass, pending the connection to the Planck mass through AP28.

## **Where the philosophical-register implementations live**

The mass derivation is read in plain-language register in Ø Predictions (Ø Models catalogue), Part I — The Proton's Weight, which constructs the same ratio by the same integer layers ( $21^2 \times 4 + \dots$ ) and the same  $\alpha$ -order dynamic corrections; the neutron-proton material (KS-NPP) is read in Part III — The Neutron's Whisper.

## **Structural relationships to subsequent work**

AP24 (The Residual), locked in Notebook IV, reads the proton mass as one of the six measured faces of  $\varepsilon$ ; AP30 supplies the value of that face from {S, B, R, C} — framework-and-value, not a cycle. AP30 opens Notebook V — Particles and Matter — which AP22 (The Ledger) and AP26 (The Surplus) complete: AP22 accounts for the antimatter by topological segregation, AP26 reads the matter surplus as the energy of the break.

## **The 420 Code placement**

Notebook V — Particles and Matter, Chapter 1.

## **How the 420 Code reads now**

With AP30 in place, the body of work expresses every fundamental scale in terms of the axioms and one measured input:  $c$ ,  $\hbar$ , and  $G$  through AP03 and AP28, and now  $m_p/m_e$

through the channel geometry. The constants are no longer a list of independent numbers; they are faces and weights of one break.

## **The closing**

Mass is not stuff. It is how hard the substrate resists you.

Eighteen hundred and thirty-six times, for the proton — to five parts per billion.

# Claim Summary

**§1 Definitions [FRAMING / DEFINITION].** Mass is geometric resistance; the electron is the unit; the 420 Code numbers 21, 3, 4 and  $\alpha$  are installed.

**§2 Lemma 1 [STRUCTURAL ARGUMENT].** Independent constraints contribute additively (from axiom independence, AP20); KS-30.1 live.

**§3 Lemma 2 [DERIVATION].** The static resistance  $1764 + 63 + 9 = 1836$ , decomposed into manifold capacity, face projection, and exchange matrix; the decomposition is the unique hierarchical construction from {21, 3, 4}.

**§4 Lemma 3 [DERIVATION].** The dynamic maintenance term  $\alpha \times 21$ , the break sustaining itself against the substrate's leakage across all channels.

**§5 Lemma 4 [DERIVATION].** The isotropic-leakage correction  $(1 - 1/(84\pi))$ , the dynamic coupling radiating into 3D, normalised by the same dimensional structure that generates the static count.

**§6 Proposition [DERIVATION].**  $m_p/m_e = 1836 + \alpha \times 21 \times (1 - 1/(84\pi)) + O(\alpha^2)$ : the  $O(\alpha)$  term gives 5 ppb with zero free parameters; the  $O(\alpha^2)$  term reduces the residual to  $\sim 0.008$  ppb (1836.15267344 vs measured 1836.15267343) using a coefficient whose structural derivation is owed under KS-30.3.

**§7 The Structural Residual [SYNTHESIS].** The residual is  $\varepsilon$  itself — the break persisting at the bottom of every order, exactly as the axiom  $1:1 + 1 \times \varepsilon$  @ AS demands.

**§8 Connections [INTEGRATION].** Ties the result to AP06, AP10, AP19, AP20, AP28, AP29, AP03, and the nuclear binding-energy ceiling.



# Conditionality Footer

**Dependencies.** AP06 (The Leakage Constant), AP10 (The Dimension), AP19 (The Direction), AP20 (The Proof), AP24 (The Residual — six-face framework), AP28 (The Constant), AP29 (The Actualization Proof), AP03 (The Ratio).

**Dependents.** AP24 (The Residual) treats the proton mass as one of six measured faces of  $\varepsilon$ , whose value AP30 now derives — framework-and-value, not a cycle; AP22 (The Ledger) and AP26 (The Surplus) build on the matter sector this paper opens.

**Kill switches closed by this paper.** None.

**Kill switches that remain live.** KS-30.1, KS-30.2, KS-30.3, KS-30.4, KS-NPP.1, KS-NPP.2, KS-NPP.3.

**Dated note, 4 September 2026.** KS-NPP.1 fired on 2 August 2026 and is shown, never repaired; the six others stand. AP47's KS-FLIP.1 to KS-FLIP.4 stand beside them (Chapter 2), with KS-30.2 cross-listed and counted once.

**Structural debts owed.** Additivity formalisation (KS-30.1); the rest of the mass spectrum and the generation structure (meets KS-66 in AP27); the absolute electron-mass scale.

**Items held open without claiming debt status.** Closed forms for  $O(\alpha^2)$  and higher (KS-30.3); the absolute electron-mass energy scale, pending AP28.





## Chapter 2

# The Flip

*The matter-side cost of a held distinction — the neutron's realised difference at second order, from the family's one mechanism.*

*Artist's Proof 47*

## Artist's Note

This is Artist's Proof 47 of The 420 Code. It belongs to Notebook V — Particles and Matter, beside AP30's proton, and it prices what the old formula priced as free.

Structurally perfect balance is impossible. Always. The reason why anything is, is the minimal structural imbalance — that is the key to the entire 420 Code. A very small imbalance is structural; it has to be. In AP40 — The Irrational, this structural imperfection, or imbalance, is conceptually developed.

The neutron is where that structural imbalance meets a scale, at the seventh decimal place, and either survives or dies there.

On the 2nd of August 2026 my neutron prediction died — cleanly, at  $7.24\sigma$ . I made a mistake and I was wrong. That means, simply, that I betrayed the structure and did not look closely enough. The death forced me to look harder and deeper. And the structure's deepest wound became its biggest blessing in disguise.

My mistake was a structural omission. Priced at the arena, the same omission sharpened the gravity derivation — AP44, The Snap: the realised coupling of its section 5. Priced here, finding it did not revive the fired row; it produced a new one beside it. Nothing got fixed. The corpse stays.

This is living work. I will work at it until the day I die, and nothing in it is above correction — the switches below are guides for my own thinking first, and teeth for the claim second. I made a mistake. I might have made more. So what. What is important is that the system can correct itself, and it did — the switches caught the structural error; the audits caught the arithmetic. Both were needed, and neither found the other's. Now, I wait. Nature agrees, or nature doesn't. That is all that matters.

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### **Claim Summary**

## 0 — Dependency and Scope

### 0.1 — What this paper does

AP47 prices the strained hold: what a configuration pays, at second order, to keep a difference open against its ground state. Three things, each at its stated grade.

First. It derives the form and scope of the cost. A held distinction discounts the booking of the difference itself — a quotient, never a new term — and the cost is defined on flips: no flip, no cost, the ground excluded by definition.

Second. It names the cost as one process — the window's leak — with a located rate (the substrate's,  $\alpha$ , from AP30's own lemma), a located depth (the held grain's  $\alpha$ -worth, carried with its conditional), and a located geometry (the pair's eight dimensional expressions under AP30's own construction, giving  $1/(8\pi)$ ).

Third. It freezes one row against nature.  $\Delta_{\text{real}} = \Delta_{\text{bare}}/(1 + \alpha^2/8\pi) = 2.53098857035$  at CODATA-2022  $\alpha$ , against the measured  $2.530988574 \pm 7.4 \times 10^{-7}$  — a residual of  $-0.005\sigma$  where the bare row fired at  $7.24\sigma$  — with the switches of section 10 armed on it.

It also rules the bound neutron (the distinction is held against the structural alternative; binding closes the exit, not the

window) and declares, rather than dresses, that the ruling carries no mass-face observable.

## 0.2 — Dependencies

AP30 (The Resistance): the dimensional-expression construction and its leakage lemma — the static count through the four spacetime dimensions, the isotropic  $1/(\text{count} \cdot \pi)$  normalisation, the substrate's rate ("the substrate speaks at  $\alpha_{\text{em}}$ "), and the principle quoted at its own hedge: *"the leakage is normalised by the very structure it leaks through."* AP46 (The Stretch, v0.2): the mechanism in its exact words — *"holding is tension, and tension sheds"* — the pairs-balance ruling, and the thread ruling that fixes the family's one base rate. AP45 (The Blink, draft v0.2.1 — the version of record by the author's word; a draft awaiting his lock, and every inheritance from it says so): the toll's two faces, the spent clause, the no-new-energy-term refusal, the grain's denomination. AP44 (The Snap, locked): the  $\varepsilon = \alpha_{\text{em}}$  selection, argued in its section 5, booked as D-CCC.4. AP06 (The Leakage Constant, published corpus): the domain field the selection was made among, held open at its section 0.7. WP-ROAD.1 v1.2: the family instrument. One measured comparator, CODATA 2022, lines named in section 6. No other inputs exist.

### 0.3 — Claimed, and refused

Claimed: that a configuration holding a distinction — an orientation differing from its ground state under the same grip — pays a continuous second-order cost on the booking of the difference itself; that the cost is one process, the window's leak, with a located rate, depth, and geometry; that its order is  $\alpha^2$  (the grain's worth times the substrate's rate) at the structural register, riders stated; that its divisor is  $8\pi$ , derived by AP30's own construction at support two; that the realised difference is the expression  $\Delta_{\text{bare}}/(1 + \alpha^2/8\pi)$  at CODATA-2022  $\alpha$ ; that the bound neutron pays; and that the binding ruling carries no mass-face observable, under either answer, declared rather than dressed as an edge.

Refused, in advance: any new term in any energy budget at any size; any identification of  $\Delta_{\text{bare}}$ 's terms with the standard electromagnetic/strong decomposition; any repair of the fired row KS-NPP.1; any third-order term; any claim on the bound neutron's mass face; any probability deviation; and any contact between the construction and standing measured numbers beyond those disclosed in this paper — the eliminations of section 5 and the proton check of section 7, where data killed candidates and generated none, and the single frozen comparison of section 6 — all made after the structure was described, in the order the record shows.

## 0.4 — The honesty box

The bare row fired at  $7.24\sigma$  and is shown in full in section 7; this paper prices the structure the fired row was missing and never touches the corpse. The paper's central structural claim — AP30's maintenance mechanics read at a strained hold instead of a ground's upkeep — is STATED, located, and switch-guarded, not derived from anything deeper; every grade in this paper rides it in the open. The order carries three riders (T-b, a registered conjecture; D-CCC.1, the formalisation; the  $\varepsilon = \alpha_{em}$  closure, DERIVED-CONDITIONAL-ON-AP45 via D-CCC.4) and the divisor inherits KS-30.2's isotropy assumption, live in its parent. The landing itself, at  $-0.005\sigma$ , is weak evidence by the family's own standard: the paper is worth having for the structure and the falsification surface, not for the number.

## 0.5 — Epistemic status per section

Section 1: the family seat, orientation. Section 2: DEFINITIONAL — the scope, ruled. Section 3: DERIVED-AT-THE-STRUCTURAL-REGISTER — the mechanism, inherited and read on the mass face. Section 4: the process STATED and located; the order  $\alpha^2$  DERIVED-AT-THE-STRUCTURAL-REGISTER with riders. Section 5: the divisor  $8\pi$  DERIVED, wearing its clauses and inheritance. Section 6: EMPIRICAL — one frozen row, reported once, never evidence of the structure.

Section 7: deaths, shown. Section 8: RULED, with a declared null. Section 9: the census, open and honest about its size.

## **0.6 — Kill switch summary**

KS-FLIP.1 — the frozen row dies if measurement departs beyond the bar. KS-FLIP.2 — forced structure; fires on any element shown selected by comparison with measurement; binding on author and desks. KS-FLIP.3 — the truncation; the absent  $\alpha^3$  term is a prediction, dated at  $\sim 20\times$  better B\_d metrology. KS-FLIP.4 — the scope; dies if a ground state requires the cost or a domain member escapes it. KS-30.2 — inherited isotropy; both divisors move together. KS-NPP.1 — fired 2 August 2026, shown forever. Full statements in section 10.

## **0.7 — Items held open without claiming debt status**

The lifetime dynamics of the window's snap — the free neutron's decay clock, including the standing beam-versus-bottle tension in its measurement — forward-register, no claim made. The census of section 9. The  $\alpha^3$  silence until metrology reaches it. The ROAD v1.2's AP45 state-table line, corrected at that instrument's next revision (a v0.3 is recorded there that does not exist in the author's archive; v0.2.1 is the version of record by his word; should a file answering that entry ever

surface, its four fence clauses are re-read before anything relies on it).

## 0.8 — Structural relationships

AP30: parent of the construction; KS-30.2 shared; no locked value touched (WP-FLIP.1 section 9's independence sentence superseded as stated in section 5). AP46: source of the mechanism and the pairs-balance ruling, both quoted verbatim; the one-base-rate reconciliation runs on its own thread ruling. AP45 (draft): source of the toll clauses and the grain's denomination; every inheritance conditional on its lock. AP44: source of the  $\varepsilon = \alpha_{em}$  selection; D-CCC.4 carries the closure. AP06: the domain field. AP40: the conceptual development of the structural imbalance this paper prices. AP42/AP28 and the cosmology: untouched. The Assembly (AP48, forthcoming): none of its content is load-bearing here, and none of this paper's rows move on its verdict. One line is filed to the record-algebra programme (D-CCC.1): the availability field recomputes structurally, blind to energetic accessibility — section 8's ruling is an axiom of that algebra. No new debt books: D-FLIP.1 was reserved for an underived divisor and is not needed.

## 0.9 — Symbols and numbers

$\Delta_{bare} = 3(1 - 1/2\pi) + \alpha(1 + 1/2\pi)$ : the ideal neutron-proton difference in electron masses, the flip priced as free.  $\Delta_{real} =$



$\Delta_{\text{bare}}/(1 + \alpha^2/8\pi)$ : the realised difference, the claim of this paper.  $f_2 = \alpha^2/8\pi$ : the hold-cost fraction.  $\alpha = 7.2973525643(11) \times 10^{-3}$ , CODATA 2022.  $\Delta_{\text{meas}} = 2.530988574$ ,  $u = 7.4 \times 10^{-7}$  (0.29 ppm) — CODATA 2022, the exact difference of  $m_n/m_e = 1838.68366200(74)$  and  $m_p/m_e = 1836.152673426(32)$ , the uncertainty stated decade-safe because its parenthetical form caused one audited slip. **The window:** the held gap between a coupling's actual and ground orientations. **The window's leak:** the one process of section 4.  **$\varepsilon$ , three postures:** the arena's splinter (held forever), the Blink's slivers (spent per event), the flip's grain (held-and-leaking, terminally released at the snap-shut).

# 1 — The Family Seat

Four costs, one axiom, one mechanism read at four sites. The Snap prices existence, once, at the arena: the held  $+1 \times \varepsilon$ , the first-order discount  $1/(1+\alpha)$  on the coupling. The Blink prices every actualization: one grain of possibility, dimensionless, the alternatives not taken. The Stretch prices persistence: the tension's continuous shed, the sky's own growth. The Flip prices the strained hold: what a configuration pays, at second order, to keep a difference open against its ground state.

The ladder, stated carefully because canon itself sharpens it. The first-order cost is the arena's, booked once, on the coupling — it is not the proton's, and no reader should hunt for it inside 1836. Ground states carry the costs of their own constitution and upkeep: AP30's proton carries a first-order maintenance leakage,  $1/(84\pi)$  on its dynamic term, and a second-order recursion — the leakage of the leakage, the break operating on its own maintenance — whose coefficient is carried with its structural decomposition owed under KS-30.3. What a ground state does **not** carry is a distinction. It holds nothing open against itself, and so it pays no distinction cost at any order. The Flip is the distinction cost: defined on flips, and only there.

## 2 — What Is Held

A ground state is a tug-of-war settled at the rope's middle: the couplings sit where the balance rests. The proton is that state — three couplings in the minimal orientation, nothing extra, nothing held open.

The neutron is not the proton plus a thing. Same inventory: three couplings, nothing added. One of the three sits in the other orientation. What the neutron holds is therefore not an object but a **distinction** — the gap between where one coupling sits and where the balance would put it. A window held open against the wind that wants it shut.

That is the whole scope, and it is definitional: the cost is defined on flips. No flip, no cost. The proton is excluded not by a check but by the definition — there is no distinction to hold when you are the ground.

The form of the correction is the fingerprint of this scope. A new thing would add a term. A held distinction discounts the difference itself — you multiply the ideal gap by the cost of keeping it open, and you touch nothing else. The old formula's omission was exactly this: it priced the flip as if holding it were free, and nothing in the ledger holds for free.

### 3 — Why a Held Distinction Costs

Holding is tension, and tension sheds — the family's derived mechanism, now cited at its verbatim locus: AP46 pays the debt in those exact words, "*holding is tension, and tension sheds*," inherited from the Blink and the Stretch and read here on the mass face. A strained hold sheds more than its ground state. The excess is spent, and spent books nowhere. So the persisting booking of the difference is discounted: the realised gap sits below the ideal one. The sign was never chosen; it is what "spent" means.

Now the step that makes the cost small, and it comes straight from the axiom. When the grip re-settles around the flip, the re-settling is free — pairs balance, and the locus is file-verified: AP46's ruled reading, verbatim, "*the pairs balance at every AS instant ... and the moving traffic leaks*." And a residual remains anyway, because **perfectly balanced is structurally impossible**: if it equalled out perfectly, there would be nothing there. The reason anything is, is the minimal structural imbalance. The axiom names the mandatory leftover:  $+1 \times \varepsilon$ . The Stretch already said this at cosmic scale, in its own words: "*the leak is the  $\varepsilon$ 's signature riding on top of a balanced flow, not the pairs failing*" — the same architecture this paper reads at one strained hold.

Singular is the load-bearing word. A dynamical failure-to-rebalance would be distributed — strain smeared across the inventory, a channel count owed somewhere. The axiom's residue is one grain, and one grain smears across nothing. Nothing in the inventory is strained. One  $\varepsilon$  is held. **The open window is the flip-site's  $\varepsilon$ .** The grain is not a fourth item in the inventory — section 2's "nothing added" stands; it is the strained configuration's own residue, held on the possibility face, which is why it can be one grain while the inventory stays three couplings. The flip's constitution is an instance of the one event class the family has confirmed at three scales — arena, actualization, persistence — and reads here at a fourth site; the held residue of that event is one grain, no matter the crowd.

Three postures of  $\varepsilon$ , so no reader conflates them: the arena's splinter, held forever, never cycling; the Blink's slivers, spent per event; and the flip's grain — **held-and-leaking** — held open across time, shedding continuously, terminally released when the window snaps shut. In a stable nucleus the window never snaps shut, and the hold sheds forever. That is not an anomaly; the Stretch sheds forever too. Spent books nowhere. That is what spent means.

## 4 — The Window's Leak: One Process

The cost is one process, and this paper names it before dividing it into an order and a divisor. **The window's leak:** the held grain leaks through the strained hold's own structure, into the substrate, at the substrate's rate. Its three parts each have an address.

**The rate is the substrate's, and the locus is canon.** AP30, Lemma 3: "the substrate's leakage rate is  $\alpha_{em}$  (AP06: the leakage constant)" — "the substrate speaks at  $\alpha_{em}$ " — and in AP30's own summary,  $\alpha_{em}$  "provides both the dynamic maintenance rate and the perturbative expansion parameter." The wind that wants the window shut is the same substrate that presses on every break; it speaks at one rate everywhere. The flip's hold sheds against it at that rate: one factor of  $\alpha$ . One voice at every door, file-verified: AP46's shed rate is "*one part in  $\alpha^{-18}$  per electron-tick*" because, by its own thread ruling, escape means "*threading eighteen sequential gates — one  $\alpha$ -transparent door each.*" The door is  $\alpha$  everywhere, and each site's structure sets only how the rate assembles: parallel channels multiply it by a count ( $\alpha \times 21$ , the proton's maintenance — AP30: "*each channel contributes  $\alpha_{em}$* "), sequential gates exponentiate it ( $\alpha^{-18}$ , the eighteen threaded faces of the cosmic shed), and a held grain multiplies rate by worth ( $\alpha^2$ , the flip). Three assemblies, one base rate. No second base rate exists anywhere in the family.

**The depth is the grain's worth.** What leaks is the held  $\varepsilon$ , one grain of  $\alpha$ -worth — AP45's own words in the draft in hand: "*one  $\varepsilon$ -grain,  $\alpha$ -worth, dimensionless*" — the rider unchanged, since a draft's wording is not a lock. The denomination has a canon face — AP30 itself writes " $\alpha_{em}$  ... identified as  $\varepsilon$  — the leakage constant, the substrate's fundamental coupling strength (AP06)" — and its forcing is carried honestly, file-verified at the canon audit:  $\varepsilon = \alpha_{em}$  in the quotient was SELECTED among AP06 section 0.7's domain readings (the AP06 file holds that field open against the canonical corpus-wide  $\varepsilon = \alpha_{em}$ , exactly as cited), the selection argued in AP44 section 5 and booked as D-CCC.4 — AP44's own section 9 saying it plainly: "*the value is not [derived], and the paper says so*" — its closure route graded DERIVED-CONDITIONAL-ON-AP45, the route's target being precisely AP06 section 0.7's held-open field. This paper inherits that conditional and says so. Worth times rate:  $\alpha^2$ .

**The geometry belongs to section 5**, where the same process is counted through the structure it leaks through. One process; section 4 is its depth, section 5 its shape.

And the cost enters lawfully. It never books: it is a dimensionless fraction of the flip's own booking, which is exactly what a possibility-side cost looks like from the energy face under the family's clause. No reference is carried anywhere — not on the energy face (that would be a booking, double-counted at order one) and not on the possibility face

(dimensionless quantities do not compose with mass). The ground exerts its pull for free; the grain is held against it; the substrate's press does the rest.

**The one claim sections 4 and 5 share, stated as itself:**

AP30's maintenance mechanics — the substrate's rate, expression-leakage, normalisation by the leaking structure — read at a strained hold instead of a ground's upkeep. That transfer is the paper's central structural claim. It is not derived from anything deeper here; it is stated, located, and switch-guarded (KS-FLIP.2, KS-30.2), and every grade in this paper rides it in the open.

Grade of the order, worn on the face: **DERIVED-AT-THE-STRUCTURAL-REGISTER**, riding — the stated transfer above; T-b (balance-impossibility, a registered conjecture); D-CCC.1 (the formalisation); and the  $\varepsilon = \alpha_{em}$  closure conditional on AP45's lock.



## 5 — The Divisor: the Same Leak, Counted

The proton's own correction carries the construction, and this paper takes it from the AP30 file directly, quoted at its own hedges. The manifold has 21 geometric channels and resolves into 4 spacetime dimensions — three spatial plus one temporal (AP10, AP20). Each channel expresses through each dimension:  $21 \times 4 = 84$  dimensional expressions. The dynamic coupling leaks isotropically into the 3D substrate, and the leakage per dimensional expression per solid angle is  $1/(84\pi)$  — the 84 in the leakage denominator being the same 84 as the static count. AP30's own sentence is the principle: **the leakage is normalised by the very structure it leaks through.**

Count the window's leak the same way. The structure the flip's leak runs through is not the arena — a distinction's static count is its **orientation pair through the four dimensions**:  $2 \times 4 = 8$  dimensional expressions, each leaking under the same lemma. The divisor is  $1/(8\pi)$ . One construction, two supports: the proton's correction is a thing held in the whole arena, support 21; the flip's is a distinction, support 2, because the window's configuration space is the pair. The support difference is the scope itself — thing versus distinction — so the divisor's derivation costs the paper one sentence it already owed.

**Why four dimensions, not three.** Because "held" is the scope's own word, and to be held is to persist. The pair's three spatial expressions are the orientation's projections; its temporal expression *is the holding* — a difference with no temporal expression is an instant, not a window, and an instant prices nothing under this paper's own definition. So the count runs through all four by the scope, not by analogy. The transfer also preserves AP30's own split exactly: there too the static count runs through four while the radiation goes into three-dimensional space's solid angle. (For the record only, and never as a reason: a three-dimensional count would give  $6\pi = 18.85$ , outside the window — but the argument above stands with the window covered, as KS-FLIP.2 requires.)

Two earlier readings retire from the derivation chain, on the record:  $8\pi$  as "four conditions times the delta's native circle," and an identification of the four dimensions with the four conditions {S, B, R, C}. The first is true arithmetic and not the forcing. The second is not in AP30 — the file says three spatial plus one temporal, and the construction never needed the identification; it was deleted, not repaired, when the file check caught it.

**Order and geometry are independent axes, and canon proves it in one paper.** AP30 carries a first-order term wearing  $\pi$  (the  $84\pi$  expression-leakage) and a second-order term wearing none (the recursion  $\alpha^2 \times 21 \times 16 / 1836$ , the leakage of the leakage, no solid angle, its coefficient owed under KS-

30.3). The  $\pi$  tracks *geometric expression-leakage*, not the power of  $\alpha$ . The arena's  $1/(1+\alpha)$  is first order without  $\pi$ ; AP30's maintenance is first order with  $\pi$ ; AP30's recursion is second order without  $\pi$ ; the Flip is second order with  $\pi$  — an expression-leakage at ledger depth two. No canon rule stands against it, because canon's own pair shows the axes come apart. This paragraph was written before the grade was decided, as review demanded.

**Inheritance, on the face.** Lemma 4's leakage is assumed perfectly isotropic, and AP30's KS-30.2 is live on exactly that assumption: shown anisotropic, the  $1/(84\pi)$  correction changes — and with it, by the same lemma, the  $1/(8\pi)$ . The Flip inherits KS-30.2, and AP30's own section 8 notes its 5 ppb residual is consistent with a small anisotropic contribution, so the shared edge is genuinely live. Accordingly, WP-FLIP.1 section 9's sentence that the Flip "touches no locked value anywhere" is superseded: it still touches no locked *value* — the proton row, the G row, the clock and the cosmology stand exactly as they were — but it shares a locked paper's construction and inherits that paper's empirical switch, and says so.

Grade: **DERIVED**, wearing the support clause, the temporal-expression argument, the KS-30.2 inheritance, and the shared transfer claim of section 4. The solid-angle normalisation inside the lemma is taken on the lemma's word at both sites;

the uniformity depends only on it being identical at both, which it is by construction.

**The elimination, shown as the record of a killing.** After the structure above was described — the words first, the pair before any arithmetic — the alternatives were run against the number. Required divisor: 25.1499. Window at the audited bar ( $7.4\times10^{-7}$  m<sub>e</sub>, 0.29 ppm): [22.096, 29.183].

| Candidate               | Assembly                                   | Value  | Stand                             |
|-------------------------|--------------------------------------------|--------|-----------------------------------|
| support 2               | $2 \times 4 \times \pi$ (the construction) | 25.13  | lands                             |
| support 3               | $3 \times 4 \times \pi$                    | 37.70  | dead                              |
| support 6               | $6 \times 4 \times \pi$                    | 75.40  | dead                              |
| support 18              | $18 \times 4 \times \pi$                   | 226.19 | dead                              |
| support 21              | $21 \times 4 \times \pi$                   | 263.89 | dead                              |
| three-dimensional count | $2 \times 3 \times \pi$                    | 18.85  | outside<br>structure<br>(section) |
| $9\pi$                  | $18 \times \pi/2$ (no construction)        | 28.27  | in-win<br>dead<br>half-c<br>suppe |

|  |  |  |                       |
|--|--|--|-----------------------|
|  |  |  | singular<br>the eight |
|--|--|--|-----------------------|

The constructed assembly forces the support into [1.76, 2.32]: only 2 survives. Metrology also holds a dated tiebreak it does not currently need: roughly a threefold improvement in the bar separates  $8\pi$  from  $9\pi$  at  $\sim 2.4\sigma$ . Data eliminated candidates here; data generated nothing. The 2 was described before it was counted, and the record shows the order.

## 6 — The Application, Once

The cost is a quotient on the flip's own booking, because the quotient is what the axiom writes; the quotient and the subtractive form differ by  $1.14 \times 10^{-11}$  — a sixty-five-thousandth of the measurement bar — so the choice of form is structural, not numerical.

$$\Delta_{\text{real}} = \Delta_{\text{bare}} / (1 + \alpha^2/8\pi) = \mathbf{2.53098857035}$$

The claim is the expression, frozen at CODATA-2022  $\alpha$ , displayed at eleven decimals; KS-FLIP.1's kill test runs against the expression, not the display (the author's ruling, applied). The display's trailing decimals carry the expression, not the measurement: the bar sits at the seventh decimal and nature adjudicates there; digits beyond it exist for exact reproducibility of the claim and carry no metrological weight.

Against the measured line, stated decade-safe:  $\Delta_{\text{meas}} = \mathbf{2.530988574}$ ,  $u = \mathbf{7.4 \times 10^{-7}}$  (**0.29 ppm**) — CODATA 2022, from  $m_n/m_e = 1838.68366200(74)$  and  $m_p/m_e = 1836.152673426(32)$ ; the central value is the exact difference of those two lines. The residual is  $-0.005\sigma$  where the bare row fired at  $7.24\sigma$ .  $\alpha$  is CODATA 2022,  $7.2973525643(11) \times 10^{-3}$ . At ten decimals the row is vintage-insensitive; at the eleven-decimal display it is not — CODATA-2018  $\alpha$  gives  $2.53098857036$ , one in the last place. The freeze therefore names the  $\alpha$  line, and the kill test runs against the expression,

not the display. The realised row enters the registry as a new frozen claim beside the corpse, never in its place.

**What the claim is, and is not.** The claim is the *total*:  $\Delta_{\text{bare}}$ 's two terms are structural layers of the corpus's own construction, and this paper identifies neither of them with the standard decomposition into electromagnetic and strong contributions. That decomposition assigns the electromagnetic piece roughly minus one MeV — opposite in sign to the  $\alpha$ -term here — and the flip cost, at about 2.7 eV, is five orders of magnitude too small to be a repair of any such piece, and is not one. Anyone attacking the anatomy is attacking a claim this paper does not make; the claim that can die is the total, at the frozen row.

## 7 — The Deaths, Shown

KS-NPP.1 fired at  $7.24\sigma$  on 2 August 2026, on the bare difference — and the source audit confirms the frozen figure on the audited bar ( $7.235\sigma$ ). It is shown here in full, it goes nowhere, and it will always be shown. No repair was offered then and none is offered now: this paper does not fix the fired row, it prices the structure the fired row was missing, and the new row stands beside the corpse. The proton check is also shown as the lawful elimination it was: applying the flip cost to the ground state moves the proton by  $0.00389$  electron masses — **121,576 $\sigma$**  at the named bar,  $u(m_p/m_e) = 3.2 \times 10^{-8}$ , CODATA 2022 — and kills it instantly. (The origin note quoted  $223\sigma$  against an unnamed coarser bar; the honest figure only strengthens the elimination.) Which is the definition doing its work, since there is no distinction to hold when you are the ground.



## 8 — Binding: the Ruling

A bound neutron is still odd. The orientation is intact; nothing in the nuclear well touches the flip. The ruling, spoken and standing: **the distinction is held against the structural alternative; binding closes the exit, not the window.** The  $\varepsilon$  is a structural residue, not a dynamical option — availability in the ledger is structural, and reachability is the exit's business. The bound neutron pays. Its stability needs no second mechanism: the release is blocked (for the deuteron, energetically — there is no bound diproton; Pauli blocking proper does the work in larger nuclei), and a blocked exit over an open window is a hold that sheds forever, which the family already knows how to say.

**Proposition (the capture-chain identity).** The gamma plus recoil in  $n + p \rightarrow d + \gamma$  equals the free neutron's booking plus the proton's minus the deuteron's, identically, by energy conservation, whatever happens inside. Therefore the tabulated neutron mass is the free booking always; the frozen row of section 6 tests the free neutron regardless of this ruling; an independent free-neutron mass would match the table either way; and a closed window's 2.7 eV would hide inside the deuteron's binding bookkeeping —  $B_d = 2,224,566.362 \pm 0.373$  eV from the CODATA-2022 mass triplet — where theory is a thousand times too coarse to look.

Declared plainly, because honesty counts edges rather than collecting them: **this ruling carries no mass-face observable, under either answer.** No switch is armed on it because no measurement can reach it. The only conceivable future handle is dynamics — the lifetime physics of the window's snap — and that is forward-register, held open in section 0.7.

## 9 — What Else Flips

The census is open and currently honest about its size: the neutron is the one priced member of the domain. Other strained holds the corpus can reach will be described first and priced after, one system at a time, on the author's clock, none hunted. Every system that pays becomes a prediction; every system that structurally cannot pay becomes a consistency check; and if the neutron remains the only member, this paper stands as a one-row claim with its switches, which is enough.

## 10 — Kill Switches

**KS-FLIP.1 — the frozen row. ARMED.** The claim is the expression  $\Delta_{\text{real}} = \Delta_{\text{bare}}/(1 + \alpha^2/8\pi)$  at CODATA-2022  $\alpha$ , displayed 2.53098857035. Dies if an improved measurement of the neutron-proton mass difference departs from it beyond the bar — the bar being the audited  $1\sigma$  of the source lines:  $7.4 \times 10^{-7} m_e$  (0.29 ppm), CODATA 2022. Nature-facing, live from this lock.

**KS-FLIP.2 — forced structure. ARMED, binding on author and desks.** Fires if any element of the construction — the order, the divisor, the scope, the form, the process claim of section 4 — is ever shown to have been selected by comparison with measurement rather than by the stated structural argument. The chain's true order is on the record: the structure in words first; the pair described before any count arithmetic; the elimination arriving afterward, as elimination; every canon locus checked against its file.

**KS-FLIP.3 — the truncation. ARMED.** The construction terminates at second order: no  $\alpha^3$  term exists. The absent third-order piece sits at  $3.9 \times 10^{-8}$  — 5.3% of the audited bar, 0.020 eV — so adjudication waits on roughly twentyfold better metrology on the deuteron binding energy ( $B_d$  currently  $\pm 0.373$  eV, CODATA 2022). The claim's silence at third order is itself a prediction, and nature will eventually rule on it.

**KS-FLIP.4 — the scope. ARMED.** Formal domain: the cost operator's domain is the set of configurations holding a non-zero second-scale  $\varepsilon$  — an orientation differing from its ground state under the same grip — and ground states are outside the domain by construction. The domain is blind to energetic accessibility, per section 8. Dies if a ground state is ever shown to require the cost, or a configuration inside the domain shown to escape it.

**KS-30.2 — inherited, live in its parent.** The isotropy assumption under both  $84\pi$  and  $8\pi$ . Shown anisotropic, both divisors move together, and this paper's row moves with its parent's.

**KS-NPP.1 — fired, and shown. Forever.**

## 11 — What This Establishes and What Remains Open

**Grades.** Scope: DEFINITIONAL (ruled). Mechanism: DERIVED-AT-THE-STRUCTURAL-REGISTER, inheriting the Blink's theorem and the Stretch's tension mechanism. The transfer: STATED — located, guarded by KS-FLIP.2 and KS-30.2, ridden by everything below. Order  $\alpha^2$ : DERIVED-AT-THE-STRUCTURAL-REGISTER with its riders — the transfer; T-b; D-CCC.1; the  $\varepsilon = \alpha_{em}$  closure DERIVED-CONDITIONAL-ON-AP45 (D-CCC.4). Divisor  $8\pi$ : DERIVED, wearing the support clause, the temporal-expression argument, KS-30.2, and the transfer. The landing ( $-0.005\sigma$ ): EMPIRICAL, reported once, never evidence of the structure. Binding: RULED, with a declared null.

**Provenance and lineage.** Words first, number once: the structure was stated in neutron.docx before any of this; the pair was described before any count arithmetic; the eliminations arrived afterward, as eliminations; every canon locus was opened. Three audits closed before this conversion: the source audit (every measured number to its named CODATA-2022 line, including the resolution of the (74) decade slip at its source), the second numeric pass over the draft's own sentences, and the canon audit (every citation against its file's version of record). The corrections ledger for conversion day stands beside the frozen instruments, never

inside them: the (74),  $72.4\sigma$ , and  $223\sigma$  lines in neutron.docx; WP-FLIP.1's section 1 (74) and its superseded sentences; the ROAD's AP45 line. Lineage: draft v0.1 → v0.7, 1 September 2026, all retained; this paper carries v0.7's verified passages verbatim. FINAL v1 → v2, same day, before anchoring: section 0.3's refusal clause corrected to enumerate the paper's three disclosed contacts — a front-matter statement the body's lawful eliminations contradicted, caught by review. FINAL v2 → v3, same day: the Artist's Note corrected on review — the recovery wording replaced (a new row beside the corpse, never a revival, the author's stated intention), the gravity claim located to AP44 section 5, and the self-correction sentence split into what the switches caught and what the audits caught; the Note's register kept as the author ruled it. The fired row stays on the board. The corpse is shown.

**The closing.** One structural sentence — a held distinction pays to stay held — priced by the family's one mechanism at its fourth site, landing on the neutron at the seventh decimal place. If the row dies, the paper dies with it, and the corpse will be shown here.

## Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies, claims and refusals, the honesty box, epistemic status, switch summary, items held open, relationships, symbols.

Section 1 (The Family Seat). ORIENTATION. Four costs, one axiom, one mechanism at four sites; the ladder stated so no reader hunts the arena's cost inside 1836.

Section 2 (What Is Held). DEFINITIONAL. The neutron holds a distinction, not a thing; no flip, no cost; the multiplicative form as the scope's fingerprint.

Section 3 (Why a Held Distinction Costs). DERIVED-AT-THE-STRUCTURAL-REGISTER. Holding is tension and tension sheds (AP46, verbatim); pairs balance and the singular  $\varepsilon$  remains; the open window is the flip-site's grain; three postures of  $\varepsilon$ .

Section 4 (The Window's Leak). PROCESS STATED; ORDER AT THE STRUCTURAL REGISTER. One process with a located rate (the substrate's  $\alpha$ ), depth (the grain's  $\alpha$ -worth, conditional carried), and the transfer claim stated as itself. Worth times rate:  $\alpha^2$ .

Section 5 (The Divisor). DERIVED. AP30's construction at support two:  $2 \times 4 = 8$  expressions,  $1/(8\pi)$ ; the temporal



expression is the holding; order and geometry independent axes; KS-30.2 inherited; the elimination shown as a killing.

Section 6 (The Application). EMPIRICAL.  $\Delta_{\text{real}} = \Delta_{\text{bare}}/(1 + \alpha^2/8\pi) = 2.53098857035$  at CODATA-2022  $\alpha$ , against  $2.530988574 \pm 7.4 \times 10^{-7}$ :  $-0.005\sigma$ . The claim is the total, never the anatomy.

Section 7 (The Deaths). SHOWN. KS-NPP.1 at  $7.24\sigma$ , forever; the proton eliminated at  $121,576\sigma$ .

Section 8 (Binding). RULED. The exit closes, not the window; the bound neutron pays; the capture-chain identity; no mass-face observable, declared.

Section 9 (The Census). OPEN. One priced member, honestly counted.

Section 10 (Kill Switches). Four armed, one inherited, one fired and shown.

Section 11 (Establishes and Open). Grades with riders on the face; provenance in the true order; the closing.

*This work is published for free, forever — at the author's word.  
Keep the signal clean.*

**Items held open without claiming debt status.** Closed forms for  $O(\alpha^2)$  and higher (KS-30.3); the absolute electron-mass energy scale, pending AP28.



## **Chapter 3**

# **The Ledger**

*Antimatter segregated behind every horizon*

*Artist's Proof 22*

# Artist's Note

*What this paper is, and why.*

You exist because of an imbalance so slight it should not have mattered. For every billion particles of antimatter the early universe made, it made a billion and one of matter. That one extra part is the entire visible universe.

This paper does not hunt for the process that made the extra. It says the ledger always balanced. The antimatter did not vanish — it was folded. Every event horizon is a mirror that swaps matter for antimatter, left for right, forward for backward. What looks like matter from our side looks like antimatter from the other. The books balance across the horizon; we have been reading one column and calling the other a mystery.

The mechanism is a theorem. The event horizon is the  $\sigma$ -boundary built into Axiom S, and at that boundary the involution is the Tomita–Takesaki modular conjugation — proved unique, proved anti-linear, proved equal to CPT. The machinery is standard algebraic quantum field theory; the identification is the new thing.

And the door stays open because the break  $\varepsilon$  has no mirror. The axiom  $1:1 + 1 \times \varepsilon @ AS$  has one uncancellable entry, the

splinter with no  $\sigma$ -image. That is why there is something rather than nothing.

AP22 derives the mechanism, not the magnitude. The numerical value of the asymmetry,  $\eta \approx 6 \times 10^{-10}$ , is owed — a debt tracked across the body of work as KS-60 and discharged in form by AP26 (The Surplus), the next chapter. The paper also carries a quantitative tension with teeth: KS-47 asks whether the mass behind all horizons can account for the missing antimatter. The blade is named.

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# Orientation

## Where this paper fits in the body of work

AP22 is the second chapter of Notebook V — Particles and Matter. AP30 (The Resistance) derived the mass of matter; AP22 accounts for the antimatter; AP26 (The Surplus) reads the leftover. The three are one argument: AP30 weighs the matter, AP22 says where its antimatter went, AP26 says how much was left. AP22 reads after AP08 (which gives the Schwarzschild horizon) and AP17 (which gives the Eye — the 1-pole / 0-pole structure the horizon expresses).

## How to read this paper

The spine is Theorem 1, proved in two parts across §4. Part A locates the  $\sigma$ -boundary at the event horizon, by geometry (AP17). Part B identifies the involution with the modular conjugation, by algebra (Tomita–Takesaki, Sewell). §4 is the dense core; its gap assessment tags every soft joint honestly. §5 reads the consequence — segregation, and why the door stays open. The connective passages speak to you directly; the theorem speaks in the precision algebraic quantum field theory requires. The gear-changes are intentional.

## 0 — Dependency and Scope

### 0.1 What this paper does

This paper derives the structural mechanism of baryogenesis from  $\{S, B, R, C\}$ .

The observed matter–antimatter asymmetry arises from topological segregation at the actualisation event: matter propagates outward (the  $\mathcal{L}$ -sector, 1-poles), antimatter collapses inward (the  $\mathcal{P}$ -sector, 0-poles, the interior of event horizons).

The net asymmetry is held open by the break  $\varepsilon$ , which has no  $\sigma$ -image (Proposition 1, unconditional).

The Horizon Conjugation Theorem (§4) derives the identification of the event horizon as the  $\sigma$ -boundary from the axioms, via algebraic quantum field theory (AQFT) and the Bisognano–Wichmann / Sewell modular conjugation theorem.

The two-sector topology (§4.2) resolves the Kruskal region mismatch. Five tagged gaps remain (§4.4); KS-46 is upgraded from LIVE—HARD to ADDRESSED.

The paper does not derive the numerical value of the baryon asymmetry ( $\eta \approx 6 \times 10^{-10}$ ). This debt is tracked across the body of work as KS-60, owned by AP26 (The Surplus).

## 0.2 Dependencies

The dependency chain: Axiom S (Symmetry —  $\sigma$ -involution, two sectors)  $\rightarrow$  Axiom B (Break —  $\varepsilon$  has no  $\sigma$ -image)  $\rightarrow$  AP17 (The Room: 0-pole/1-pole structure, the Eye)  $\rightarrow$  AP08 (Einstein field equations, Schwarzschild solution)  $\rightarrow$  this paper (Theorem 1: event horizon =  $\sigma$ -boundary;  $\sigma^{\wedge}$  = CPT at the horizon).

**Depends on:** Axiom S (Symmetry —  $\sigma$ -involution), Axiom B (Break —  $\varepsilon$  has no  $\sigma$ -image), Axiom R (Record — irreversibility; kills white holes), Axiom C (Constraint; forces fold; microcausality), AP05 (Lorentzian spacetime; Poincaré covariance), AP09 (complex Hilbert space), AP08 (Einstein field equations; Schwarzschild solution), AP17 (the Eye; 1-pole/0-pole), AP20 (AS = manifold), AP21 (Steiner tree; Energy-Measure Bridge).

## 0.3 Axiom mapping

Axiom S supplies the involution  $\sigma$  (two sectors,  $\mathcal{L} \leftrightarrow \mathcal{P}$ ) and, with it, the  $\sigma$ -boundary. Axiom B supplies the break  $\varepsilon$ , which has no  $\sigma$ -image — the uncancellable entry that keeps the ledger from closing. Axiom R kills the white-hole region (records cannot be unwritten), collapsing the four-region Kruskal structure. Axiom C forces the fold and bounds propagation at  $c$ , fixing the causal structure in which the event horizon is defined.



## 0.4 Epistemic status per section

**Epistemic status per section.** §1 (Crisis of Asymmetry): historical — summary of the baryogenesis problem and Sakharov conditions. §2 (Involution and the Eye): established — summarises Axiom S and AP17. §3 (Two Kinds of Antimatter): derived — local antimatter from pair production is standard; distinction from net asymmetry follows from  $\sigma$ -structure. §4 (Horizon Conjugation): derived — Theorem 1 proved via Tomita–Takesaki modular theory and the Bisognano–Wichmann / Sewell line, with the two-sector topology resolving the Kruskal region mismatch; five tagged gaps (§4.4), all SMALL or smaller, with Gap 5 CLOSED. §5.1 (Segregation): derived — matter outward / antimatter inward follows from Theorem 1 + AP17. §5.2 (Door Stays Open): derived —  $\varepsilon$  has no  $\sigma$ -image, from the axiom directly, formalised as Proposition 1, independent of Theorem 1. §5.3 (Ratio): debt — structural reason derived; numerical magnitude owed (tracked as KS-60, AP26). §6 (Formal Mapping): derived — quantum-number balance follows from Theorem 1, formalised as Proposition 2.

## 0.5 Kill switch summary

Five kill switches engage on this paper. KS-46 (black-hole conjugation) is ADDRESSED, split into KS-46A (boundary identification, DERIVED), KS-46B (AQFT bridge, ADDRESSED — bifurcation tension discharged via Sewell 1982, broader

bridge SMALL gaps remaining), and KS-46C (operator identification, ADDRESSED). KS-47 (global baryon number) and KS-53 (Hawking evaporation products) are LIVE — EMPIRICAL: both are quantitative predictions with teeth.

## **0.6 Structural debts owed**

The numerical value of the baryon asymmetry,  $\eta \approx 6 \times 10^{-10}$ , is owed. This paper derives the structural reason for the asymmetry ( $\varepsilon$  has no  $\sigma$ -image); the magnitude is a calculation, tracked across the body of work as KS-60 and discharged in structural form by AP26 (The Surplus). The mass-balance question — whether the mass inside all horizons can account for the missing antibaryonic content — is carried as KS-47.

## **0.7 Items held open without claiming debt status**

The five gaps tagged in §4.4 (AQFT framework adaptation, anti-linearity construction on the specific algebra, the Hartle–Hawking /  $\sigma$ -invariance identification, rotating black holes, and the bifurcation tension — the last now closed via Sewell 1982). The remaining gaps are all SMALL.

## **0.8 Structural relationships**

AP22 reads after AP17 (the Eye), AP08 (the Schwarzschild horizon), AP21 (the Energy–Measure Bridge and filament fragmentation), and AP20 (AS = manifold). It opens onto AP26

(The Surplus), which reads the same surplus as the energy of the break,  $\eta = E(\varepsilon)$ ; the two papers are complementary — AP22 gives the mechanism (where the antimatter is), AP26 gives the form (how much matter is left), reconciled by the cross-horizon balance: the surplus on our side is matched by an equal and opposite surplus on the other, total exactly zero.

## Notation

$\varepsilon$  — the break. Minimum viable splinter. Always Axiom B. The only fundamentally asymmetric object in the argument.

$\sigma$  — the involution mapping  $\mathcal{L} \leftrightarrow \mathcal{P}$  (Axiom S). On the manifold, maps matter  $\leftrightarrow$  antimatter.

$\mathcal{L}, \mathcal{P}$  — the two sectors of the pre-state.  $\mathcal{L} \rightarrow$  exterior manifold (propagation, 1-poles, matter).  $\mathcal{P} \rightarrow$  interior of horizons (fold, 0-poles, antimatter).

**1-pole** — propagation, matter, the outward condition. Light. The white of the Eye (AP17).

**0-pole** — fold, collapse, the inward condition. Gravity. The black of the Eye (AP17). Includes ALL gravitationally collapsed structures: SMBHs, stellar BHs, primordial BHs, any structure where Axiom C forces compactification.

$\hat{\sigma}$  — the  $\sigma$ -involution expressed as an operator on quantum states (§6). Proved equal to CPT at the horizon (Theorem 1).

**J** — the Tomita–Takesaki modular conjugation operator. The unique anti-linear involution mapping a von Neumann algebra to its commutant while preserving the cyclic vector.

**$\Delta$**  — Tomita–Takesaki modular operator. Generates the modular flow (KMS evolution at the Hawking temperature).

$\eta$  — baryon asymmetry parameter.  $\eta = (n_B - n_{\bar{B}})/n_\gamma \approx 6 \times 10^{-10}$ . Observed (Planck 2018).

$\mathcal{I}^+$  — future null infinity. The asymptotic boundary reached by outgoing light rays.

# 1 — The Crisis of Asymmetry

You exist because of an imbalance so slight it should not have mattered.

In the early universe, energy should condense into equal parts matter and antimatter. These should annihilate, leaving a universe of radiation. No structure, no chemistry, no life.

They did not. The visible universe is composed almost entirely of matter. The observed baryon asymmetry parameter is  $\eta = (n_B - n_{\bar{B}})/n_\gamma \approx 6 \times 10^{-10}$ : approximately one extra baryon per billion photons.

This tiny surplus — one part in a billion — is the entire visible universe. Everything else annihilated.

The standard approach to baryogenesis requires the three Sakharov conditions (1967): (i) baryon number violation, (ii) C and CP violation, and (iii) departure from thermal equilibrium.

The Standard Model provides some C and CP violation but not enough. No mechanism within the Standard Model produces sufficient baryon number violation. The puzzle remains open.

This paper proposes a structural resolution that sidesteps the Sakharov conditions entirely. The matter–antimatter asymmetry does not arise from dynamical processes during the evolution of the universe.

It arises from the topology of the actualisation event itself.

Note on the Sakharov conditions: The three Sakharov conditions are necessary for dynamical baryogenesis — where the asymmetry is generated during the thermal evolution of the universe.

The mechanism proposed here is topological: the asymmetry is built into the structure of the manifold at actualisation, not generated by subsequent processes. The Sakharov conditions are not violated; they are inapplicable.

If a future experiment demonstrated that baryogenesis must occur dynamically and could not be topological, this would not falsify the axioms but would render this particular mechanism inapplicable.

## 2 — The Involution and the Eye

Axiom S defines the pre-state as two sectors,  $\mathcal{L}$  and  $\mathcal{P}$ , perfectly mapped by the involution  $\sigma$ . Perfect symmetry: 1:1.

On the macroscopic manifold, this topology expresses itself as the Eye (AP17). The 1-pole is propagation, light, the outward expansion — the white of the Eye.

The 0-pole is the fold, collapse, the throat of the Eye — the black of the Eye. These are not metaphors. They are the two conditions that the axiom structure imposes on the manifold.

Because  $\sigma$  is an exact involution, the global structure preserves a fundamental balance. But  $\sigma$  does not require that both sectors are visible from the same side.

It requires only that for every element in  $\mathcal{L}$ , there exists a  $\sigma$ -image in  $\mathcal{P}$ . It does not require that the image is reachable by an observer in  $\mathcal{L}$ .

This is the key structural observation:  $\sigma$  guarantees the existence of the image. It does not guarantee the image is accessible from the same region of the manifold.

You have looked in a mirror. The reflection is there. But you cannot reach through the glass and touch it. The  $\sigma$ -involution guarantees the reflection exists.



It does not guarantee you can cross to the other side.

### 3 — Two Kinds of Antimatter

Before proceeding, a crucial distinction. Antimatter is observed in the laboratory. Positrons appear in PET scans. Antiprotons are produced at CERN. Antimatter exists in the visible universe. This is not in dispute.

The axioms account for this. Local antimatter arises through pair production — the break creating paired elements ( $\sigma$ -images) that exist briefly on the manifold before annihilating. This is the standard QFT process.

It operates entirely within the  $\mathcal{L}$ -sector and does not require the  $\sigma$ -boundary to be crossed.

The baryogenesis problem is not about local pair production. It is about the net asymmetry: why, after all annihilation is complete, does a surplus of matter remain in the visible universe?

Where is the missing antimatter?

The answer: it is inside the event horizons.

## 4 — The Horizon Conjugation

**Theorem 1 (Horizon Conjugation).** The event horizon is the macroscopic expression of the  $\sigma$ -boundary. The  $\sigma$ -involution, restricted to the horizon, implements CPT conjugation (up to spatial rotation about the radial axis).

Specifically:  $\hat{\sigma} = J$ , where  $J$  is the Tomita–Takesaki modular conjugation of the exterior algebra in the Hartle–Hawking vacuum, and  $J = \text{CPT}$  by the Bisognano–Wichmann theorem and its curved-spacetime generalisations (Sewell 1982, Summers–Verch).

The proof has two parts. Part A shows the  $\sigma$ -boundary coincides with the event horizon. Part B shows  $\hat{\sigma} = \text{CPT}$  there.

Between them, §4.2 addresses a critical geometric subtlety: the Bisognano–Wichmann theorem operates on the maximally extended four-region Kruskal geometry, but the axioms restrict the manifold to two sectors.

This restriction is proved (Proposition 0). Its consequence for the commutant identification (Corollary) is a structural reading the axioms motivate, not a result delivered by the imported modular theory — standard theory maps a wedge to its causal complement (the opposite exterior), and the identification of the interior as the commutant is asserted on the two-sector

topology, not derived from Sewell. This is the open step KS-46B holds (ADDRESSED, not closed).

## **4.1 — Part A: The $\sigma$ -boundary is the event horizon**

(A1) On the manifold (AP20, AS = manifold), the two sectors  $\mathcal{L}$  and  $\mathcal{P}$  must be expressed as two regions. The boundary between them is the  $\sigma$ -boundary: the surface where the sector character changes.

(A2) AP17 identifies the sector characters:  $\mathcal{L}$  is the 1-condition (propagation — signals can escape to arbitrary distance).  $\mathcal{P}$  is the 0-condition (fold — Axiom C forces compactification; signals cannot escape).

(A3) Axiom C imposes a finite propagation bound  $c$ . This creates the causal structure of the manifold (AP05, AP08). In particular, it creates surfaces beyond which no signal can propagate outward.

In coordinate-invariant terms, the event horizon is defined as the boundary of the causal past of future null infinity ( $\mathcal{I}^+$ ): the surface separating the region from which signals can reach infinity from the region where they cannot.

(A4) The 1-condition (outward propagation possible) holds in the exterior of the horizon. The 0-condition (outward

propagation impossible, fold) holds in the interior. The transition between the two conditions occurs at the horizon.

(A5) The  $\sigma$ -boundary separates the  $\mathcal{L}$ -sector (1-condition) from the  $\mathcal{P}$ -sector ( $\emptyset$ -condition). The event horizon separates the region where outward propagation is possible from the region where it is not. These are the same surface.

### **$\therefore$ The $\sigma$ -boundary is the event horizon. (Part A)**

Epistemic note: This argument uses the coordinate-invariant definition of the event horizon (boundary of causal past of  $\mathcal{I}^+$ ), not Schwarzschild coordinates. It depends on AP17's identification of 1-pole/ $\emptyset$ -pole with propagation/fold.

If AP17's identification is wrong, Part A is wrong. See Gap 5 (§4.4) for further discussion.

## **4.2 — The two-sector topology**

**The Kruskal problem.** The maximally extended Schwarzschild solution has four regions in the Kruskal–Szekeres diagram:

Region I — exterior (the observable universe outside the horizon)

Region II — black hole interior (future singularity — the collapse domain)

Region III — white hole interior (past singularity — the expansion domain)

Region IV — other exterior (a causally disconnected second exterior universe)

The Bisognano–Wichmann theorem and its curved-spacetime generalisations operate on this four-region structure. The modular conjugation  $J$  maps Region I  $\leftrightarrow$  Region IV (the two exteriors).

It does NOT map Region I  $\rightarrow$  Region II (exterior  $\rightarrow$  interior). The commutant of the exterior algebra  $\mathcal{A}_I$  is the algebra  $\mathcal{A}_{IV}$  of the other exterior, not the algebra of the interior.

AP22 requires  $\sigma$  to map Region I  $\rightarrow$  Region II (exterior  $\rightarrow$  black hole interior).

If the manifold had the full four-region Kruskal structure, the uniqueness argument would fail:  $J$  and  $\sigma$  would act on different targets.

**The resolution: the axioms restrict the manifold to two sectors, not four.**

**Proposition 0 (Two-Sector Topology).** The axioms  $\{S, B, R, C\}$  restrict the manifold to exactly two sectors:  $\mathcal{L}$  (exterior) and  $\mathcal{P}$  (interior).

Regions III and IV of the maximally extended Schwarzschild solution do not exist on the manifold.

**Proof.** The argument proceeds in three steps.

**Step 1 (Axiom R kills Region III).** Region III is the white hole interior — the time-reversed black hole. In Region III, matter emerges from a past singularity and expands outward.

This is a time-reversed process: it requires records to be unwritten, actualisations to be undone. Axiom R (irreversibility of records) prohibits this. White holes are structurally forbidden. Region III does not exist on the manifold.

**Step 2 (Axiom S kills Region IV).** Axiom S defines exactly one involution  $\sigma$  connecting exactly two sectors:  $\mathcal{L}$  and  $\mathcal{P}$ . The involution is unique — there is one  $\sigma$ , not two.

Region IV would be a second copy of  $\mathcal{L}$  — a second exterior. But  $\sigma$  connects one  $\mathcal{L}$  to one  $\mathcal{P}$ .

A second exterior would require either a second involution (violating Axiom S) or identification of Region IV with  $\mathcal{L}$  (making it the same region, not a new one).

Region IV does not exist as an independent sector on the manifold.

**Step 3 (The manifold is two-sector).** With Regions III and IV eliminated, the black hole spacetime has exactly two

regions: the exterior (Region I =  $\mathcal{L}$ ) and the interior (Region II =  $\mathcal{P}$ ).

The  $\sigma$ -boundary is the event horizon separating them.  $\square$

You have just watched two axioms — irreversibility and symmetry — kill half of general relativity's maximally extended solution. White holes die because records cannot be unwritten. Parallel universes die because the involution is unique.

What remains is what you observe: an outside and an inside, separated by a horizon.

**Corollary (Commutant identification — structural reading).** In the two-sector topology, the axioms motivate identifying the commutant of the exterior algebra  $a_{\mathcal{L}}$  with the interior algebra  $a_{\mathcal{P}} - a'_{\mathcal{L}} = a_{\mathcal{P}}$ . This identification is the structural reading the axioms support; it is not delivered outright by the imported modular theory and is held open under KS-46B (see §4.4).

**Structural reading.** The two-sector topology motivates treating the exterior and interior as a bipartite system,  $\mathcal{H} \approx \mathcal{H}_{\mathcal{L}} \otimes \mathcal{H}_{\mathcal{P}}$ , in which the commutant of  $a_{\mathcal{L}}$  is  $a_{\mathcal{P}}$  by the tensor-product commutation theorem. This bipartite factorisation is the structural assumption: the local algebras of the horizon are type III, for which no such factorisation holds



in general and the modular commutant  $a'_{\mathcal{L}}$  is the algebra of the causal complement rather than of the interior.

Identifying the two is the reading the axioms support; closing the gap requires either a split-property argument that delivers the factorisation across the horizon, or a direct construction of  $\hat{\sigma}$  on the type III algebra. This is what KS-46B holds open.

Why this resolves the Kruskal mismatch. In the maximally extended Schwarzschild,  $J$  maps  $a_I \rightarrow a_{IV}$  because the commutant of  $a_I$  is  $a_{IV}$ . In the two-sector topology, the commutant of  $a_{\mathcal{L}}$  is  $a_{\mathcal{P}}$  (the interior).

Therefore  $J$  maps  $a_{\mathcal{L}} \rightarrow a_{\mathcal{P}}$ . The modular conjugation now targets the correct region.

Note: The physical content of the two-sector topology is that black holes are formed by gravitational collapse from the exterior (AP21, direct vacuums), not extracted from an eternal Kruskal geometry.

This is consistent with astrophysical reality: real black holes form by collapse; Regions III and IV are artefacts of the maximal analytic extension, not physical regions.

See Gap 5 (§4.4) for the bifurcation tension this creates.

### 4.3 — Part B: $\sigma = \text{CPT}$ at the horizon

The derivation has four stages: (1) show the derived QFT satisfies the conditions needed for modular theory at horizons, (2) apply the Bisognano–Wichmann / Sewell / Kay–Wald results to the Killing horizon, (3) show that  $\hat{\sigma}$  satisfies the same defining properties as  $J$ , (4) use uniqueness to conclude  $\hat{\sigma} = J = \text{CPT}$ .

#### **Stage 1: The derived QFT satisfies the conditions for modular theory at horizons.**

The Bisognano–Wichmann theorem (Minkowski space) requires the Wightman axioms.

Its curved-spacetime generalisations (Sewell 1982; Kay–Wald 1991; Summers–Verch) require adapted conditions: global hyperbolicity, local commutativity (Einstein causality), existence of a suitable Hadamard/KMS state, and the cyclic/separating property for the wedge algebra.

These conditions are met by the derived QFT:

W1 — Local Lorentz covariance. The axioms derive Lorentzian spacetime (AP05) and the Einstein field equations (AP08). The Lorentzian signature gives local Lorentz structure in the tangent space at every point.

Global Poincaré symmetry holds only asymptotically; the relevant condition for curved-spacetime AQFT is local Lorentz

covariance, which is automatic from the derived metric structure. ✓

W2 — Spectral condition (energy bounded below). By AP21's Energy-Measure Bridge (Lemma): all energy is a manifestation of the break  $\varepsilon$ . The 1:1 is the zero-energy ground state.

Energy is proportional to the record measure  $\mu$ , which is non-negative (it counts actualisations). The spectrum of the Hamiltonian is bounded below by zero. ✓

W3 — Existence and uniqueness of vacuum. The vacuum is the 1:1 state — the state with no records, no actualisations, perfect symmetry. It exists by the axiom structure (the 1:1 is the starting point).

Its uniqueness follows from Axiom B:  $\varepsilon$  is the single generator of all departures from 1:1, so the path from vacuum to any excited state is through  $\varepsilon$ .

The vacuum is the unique state annihilated by all lowering operators. ✓

W4 — Local commutativity (microcausality). Axiom C bounds propagation at  $c$ . Two regions separated by a spacelike interval cannot communicate. Therefore observables localised in spacelike-separated regions commute:  $[a(O_1), a(O_2)] = 0$  for  $O_1, O_2$  spacelike.

This is microcausality, forced by Axiom C. ✓

W5 — Completeness (cyclicity of vacuum). The Reeh–Schlieder theorem follows from the spectral condition (W2) and local commutativity (W4) in the AQFT framework.

Since W1–W4 hold, the vacuum is cyclic and separating for local algebras. ✓

Caveat: The conditions above are stated in the language of the Wightman axioms for expository clarity. The rigorous curved-spacetime framework is algebraic QFT (Haag–Kastler nets with the microlocal spectrum / Hadamard condition).

The physical content is the same; the formal packaging differs. See Gap 1 (§4.4).

The derived QFT satisfies the conditions needed for modular theory at horizons. The Bisognano–Wichmann theorem and its curved-spacetime generalisations are available.

## **Stage 2: The modular conjugation at a Killing horizon.**

Background (Tomita–Takesaki theory).

For a von Neumann algebra  $\mathfrak{a}$  with a cyclic and separating vector  $|\Omega\rangle$ , there exists a unique pair  $(J, \Delta)$  where  $J$  is an anti-linear isometric involution (the modular conjugation) and  $\Delta$  is a positive self-adjoint operator (the modular operator), determined by the polar decomposition of the Tomita operator  $S: \mathfrak{a}|\Omega\rangle \mapsto \mathfrak{a}|\Omega\rangle$ .

The key properties:  $J|\Omega\rangle = |\Omega\rangle$ ,  $JaJ = a'$  ( $J$  maps the algebra to its commutant), and  $\Delta^{\{it\}}$  generates the modular automorphism group.

The Bisognano–Wichmann theorem (1975–76).

For a relativistic QFT satisfying the Wightman axioms on Minkowski space, the modular conjugation  $J$  associated with a Rindler wedge algebra and the vacuum implements CPT (up to a spatial rotation by  $\pi$  about the edge of the wedge):  $J = \Theta \cdot R_1(\pi)$ , where  $\Theta$  is the CPT operator.

The Sewell extension (1982). Sewell proved an analogue of Bisognano–Wichmann for quantum fields on spacetimes with bifurcate Killing horizons.

In the Hartle–Hawking (KMS) state at the Hawking temperature, the modular flow is the Killing flow, and the modular conjugation implements a CRT-type transformation (charge conjugation  $\times$  reflection  $\times$  time reversal) across the horizon.

This is the curved-spacetime result that identifies  $J = \text{CPT}$  at the horizon.

Kay–Wald (1991). Kay and Wald established the uniqueness and thermal (KMS) properties of states on spacetimes with bifurcate Killing horizons under wedge-reflection isometry assumptions.

Their results provide the state-theoretic foundation: the Hartle–Hawking vacuum is the unique KMS state at the Hawking temperature  $T_H = \hbar c^3 / (8\pi G M k_B)$ , and this KMS property is what makes the Tomita–Takesaki modular theory physically meaningful at the horizon.

Application to the argument. AP08 derives the Schwarzschild solution. A Schwarzschild black hole has a bifurcate Killing horizon (the bifurcation 2-sphere in the maximally extended solution).

The Hartle–Hawking vacuum is the unique KMS state at the Hawking temperature. By Stage 1, the derived QFT satisfies the conditions for modular theory.

By §4.2, the topology is two-sector, so the commutant of  $a_{\mathcal{L}}$  is  $a_{\mathcal{P}}$ .

Therefore the Sewell / Kay–Wald results apply:  $J$  maps  $a_{\mathcal{L}} \rightarrow a_{\mathcal{P}}$  and  $J = \text{CPT}$  (up to spatial rotation about the radial axis).

Note on collapse-formed black holes. The black holes in this argument are formed by gravitational collapse (AP21, direct vacuums), not extracted from an eternal Kruskal geometry.

A collapse-formed black hole has no past horizon and no bifurcation surface at early times.

However, at late times after collapse, the near-horizon geometry approaches the stationary Schwarzschild geometry

exponentially fast (the “peeling” property), and the physical Unruh state (appropriate for collapse) approaches the Hartle–Hawking KMS state in the exterior region.

Sewell’s result then applies in this late-time regime: the modular structure of the exterior algebra converges to that of the stationary case, and  $J = \text{CPT}$  holds asymptotically.

The segregation mechanism (§5) operates at late times — after the horizon has formed and the near-horizon geometry has settled — so this late-time applicability is sufficient.

### **Stage 3: $\sigma$ satisfies the defining properties of $J$ .**

(P1)  $\sigma$  is an involution.  $\sigma^2 = \text{id}$  (Axiom S). Therefore  $\sigma^2 = 1$ .  $\checkmark$

(P2)  $\sigma$  is anti-linear. By Part A,  $\sigma$  maps  $\mathcal{L}$  (exterior) to  $\mathcal{P}$  (interior) at the horizon. The Killing vector field  $\partial/\partial t$  is timelike in the exterior and spacelike in the interior.

The modular flow  $\Delta^{it}$  is generated by the Killing flow, which evolves forward in the exterior and has a qualitatively different character in the interior.

The modular conjugation  $J$ , which interchanges the algebra with its commutant, is anti-linear by the Tomita–Takesaki construction (it arises from the anti-linear Tomita operator  $S$  via polar decomposition).

Since  $\hat{\sigma}$  is to be identified with  $J$ , it must be anti-linear. More directly:  $\hat{\sigma}$  maps the exterior algebra to the interior algebra across the horizon, interchanging the two sectors.

By Wigner's theorem, any symmetry of quantum mechanics is either unitary or anti-unitary.

A symmetry that interchanges a von Neumann algebra with its commutant while preserving the vacuum in a KMS state is necessarily anti-linear — this is a structural property of the Tomita–Takesaki construction, not a separate assumption. ✓

Formal proof of anti-linearity. The Tomita operator  $S$  for the pair  $(\mathcal{A}_{\mathcal{L}}, |\Omega_{HH}\rangle)$  is defined by  $S(a|\Omega_{HH}\rangle) = a|\Omega_{HH}\rangle$  for all  $a \in \mathcal{A}_{\mathcal{L}}$ .

$S$  is anti-linear because the adjoint map  $a \mapsto a^*$  is anti-linear:  
 $S(\lambda \cdot a|\Omega_{HH}\rangle) = (\lambda a^*)|\Omega_{HH}\rangle = \lambda^* \cdot a^*|\Omega_{HH}\rangle = \lambda^* \cdot S(a|\Omega_{HH}\rangle).$

The polar decomposition  $S = J\Delta^{1/2}$  gives  $J$  anti-linear (since  $S$  is anti-linear and  $\Delta^{1/2}$  is a positive linear operator,  $J = S\Delta^{-1/2}$  inherits the anti-linearity of  $S$ ).

Now:  $\hat{\sigma}$  satisfies the same defining properties as  $J$  — it is an involution (P1), maps  $\mathcal{A}_{\mathcal{L}}$  to  $\mathcal{A}'_{\mathcal{L}}$  (P3), and preserves  $|\Omega_{HH}\rangle$  (P4).

By the uniqueness of the Tomita–Takesaki modular conjugation (Stage 4),  $\hat{\sigma} = J$ . Since  $J$  is anti-linear by construction,  $\hat{\sigma}$  is anti-linear.



The anti-linearity of  $\hat{\sigma}$  is therefore not an assumption but a consequence of the Tomita–Takesaki theorem applied to the pair  $(a_{\mathcal{L}}, |\Omega_{\text{HH}}\rangle)$ .  $\square$

(P3)  $\hat{\sigma}$  maps  $a_{\mathcal{L}}$  to  $a_{\mathcal{P}}$ . By Part A,  $\sigma$  maps  $\mathcal{L}$  (exterior) to  $\mathcal{P}$  (interior).

On the manifold,  $\hat{\sigma}$  maps local observables in the exterior to local observables in the interior:  $\hat{\sigma} a_{\mathcal{L}} \hat{\sigma} = a_{\mathcal{P}}$ . By the Corollary of §4.2,  $a_{\mathcal{P}} = a'_{\mathcal{L}}$ .

Therefore  $\hat{\sigma} a_{\mathcal{L}} \hat{\sigma} = a'_{\mathcal{L}}$ , which is the defining property of a modular conjugation.  $\checkmark$

(P4)  $\hat{\sigma}$  preserves the vacuum. The 1:1 is  $\sigma$ -invariant by construction:  $\sigma$  maps the 1:1 onto itself (perfect pairing under  $\sigma$ ).

The Hartle–Hawking vacuum  $|\Omega_{\text{HH}}\rangle$  is the thermofield double state — a maximally entangled state across the two sectors that is symmetric under exchange of the tensor factors.

This is precisely the  $\sigma$ -symmetric state: the state that looks the same from both sides of the horizon. Therefore  $\sigma|\Omega_{\text{HH}}\rangle = |\Omega_{\text{HH}}\rangle$ .  $\checkmark$

**Stage 4: Uniqueness  $\rightarrow \hat{\sigma} = J$ .**

By Tomita–Takesaki theory, the pair  $(J, \Delta)$  is uniquely determined by the pair  $(a, |\Omega\rangle)$  via the polar decomposition of the Tomita operator  $S$ .

Given the algebra  $a_{\mathcal{L}}$  and the vector  $|\Omega_{\text{HH}}\rangle$ , there is exactly one anti-linear isometric involution satisfying  $Ja_{\mathcal{L}}J = a'_{\mathcal{L}}$  and  $J|\Omega_{\text{HH}}\rangle = |\Omega_{\text{HH}}\rangle$ .

$\hat{\sigma}$  satisfies (P1)–(P4): it is an anti-linear involution that maps  $a_{\mathcal{L}}$  to  $a'_{\mathcal{L}}$  and preserves the vacuum.  $J$  satisfies the same properties and is uniquely determined.

**Therefore  $\hat{\sigma} = J$ .**

By the Sewell extension of the Bisognano–Wichmann theorem (Stage 2),  $J = \text{CPT}$  at the Killing horizon.

**Therefore  $\hat{\sigma} = \text{CPT}$  at the horizon. (Part B)**

**Interpretational note.** The statement “ $\hat{\sigma} = \text{CPT}$  at the horizon” is a statement about the modular conjugation associated with the wedge algebra and the Hartle–Hawking / KMS state.

It is an identification between operator-algebraic symmetries relating the two sector algebras. It is not a claim that an infalling proton dynamically transforms into an antiproton upon crossing the horizon.

The CPT conjugation relates the description of states as seen from the exterior to the description of states in the interior.

An infalling observer experiences nothing at the horizon — this is consistent with the equivalence principle and with Axiom C, which imposes the bound but does not create a local physical barrier.

Theorem 1 stands as a structural result, not a closed proof.  
Part A (§4.1): the  $\sigma$ -boundary is the event horizon — derived.  
Part B (§4.3):  $\sigma^\wedge = J = \text{CPT}$  at the horizon — resting on the commutant identification (Corollary), which is a structural reading held open under KS-46B, not delivered by the imported modular theory.

The two-sector topology (§4.2) ensures the modular conjugation maps exterior  $\rightarrow$  interior. Five gaps are tagged below.

You have just watched the axiom's involution — the mirror built into reality at the deepest level — express itself as CPT at the event horizon. The mirror is not a metaphor.

It is the Tomita–Takesaki modular conjugation, proved unique, proved anti-linear, proved to map matter to antimatter across the boundary between the two conditions of existence.

## **4.4 — Gap assessment**

The following gaps are tagged for completeness. The derivation is presented as a theorem with tagged gaps, not as a conjecture.

**Gap 1 (AQFT framework adaptation): SMALL.** The verification in Stage 1 uses the language of the Wightman axioms for expository clarity.

The Wightman axioms are formulated for Minkowski-space QFT (global Poincaré covariance, spectral condition tied to translations).

On curved spacetime, the rigorous framework is algebraic QFT: Haag–Kastler nets with the microlocal spectrum condition (Hadamard condition) replacing the flat-space spectral condition.

The physical conditions (energy bounded below, microcausality, cyclic/separating vacuum) transfer from the axioms to the AQFT setting. A fully rigorous treatment would restate Stage 1 in the Haag–Kastler language with the Hadamard condition explicitly verified.

The weakest point is W3 (uniqueness of vacuum): the argument from “ $\varepsilon$  is the single generator” is correct in spirit but would benefit from a formal proof that the vacuum is the unique translation-invariant state in the relevant representation.

**Gap 2 (Anti-linearity of  $\sigma$ ): SMALL.** The formal proof (Stage 3, property P2) shows that anti-linearity is not an assumption but a consequence of the Tomita–Takesaki construction: the Tomita operator  $S$  is anti-linear by definition ( $S(a|\Omega\rangle) = a|\Omega\rangle$ ),

*and the adjoint map is anti-linear);  $J$  inherits anti-linearity from  $S$  via the polar decomposition  $S = J\Delta^{1/2}$ ;  $\sigma = J$  by uniqueness (Stage 4).*

The gap is now SMALL: the formal proof is exhibited, and the only remaining refinement would be a fully rigorous construction of the Tomita operator on the specific algebra (rather than importing it from standard AQFT).

The supplementary Wigner theorem argument ( $\sigma$  preserves transition probabilities and reverses Killing time, therefore anti-unitary) provides an independent physical route.

**Gap 3 (Hartle–Hawking vacuum =  $\sigma$ -invariant state):**

**SMALL.** The identification is physically natural: the thermofield double is explicitly symmetric under exchange of the two tensor factors, which is what  $\sigma$  does.

A formal proof would show that the KMS condition at the Hawking temperature follows from the axioms (via the periodicity of Euclidean time in the Killing structure derived in AP08) and that the resulting KMS state is the unique state with the thermofield double structure.

**Gap 4 (Rotating black holes): MINOR.** The derivation is for Schwarzschild (non-rotating). Real astrophysical black holes rotate (Kerr).

The Sewell / Kay–Wald results extend to bifurcate Killing horizons in general, which includes Kerr, but the explicit verification for Kerr is not shown here.

**Gap 5 (Bifurcation tension): CLOSED (via Sewell 1982).**

Proposition 0 eliminates Regions III and IV using Axioms R and S.

The original tension: the Kay–Wald theorem (global version) assumes the full four-region Kruskal manifold, but the two-sector topology eliminates Regions III and IV.

Resolution (via Sewell 1982). The relevant theorem is Sewell (1982), which generalises Bisognano–Wichmann to curved spacetimes. Sewell’s hypotheses are weaker than Kay–Wald’s global version.

They require only: (1) globally hyperbolic spacetime, (2) a wedge region with a local bifurcate Killing horizon, (3) a KMS state with respect to the Killing flow.

The axioms satisfy all three: (1) AP05 derives globally hyperbolic Lorentzian spacetime; (2) the exterior region ( $\mathcal{L}$ -sector) is a wedge bounded by a local bifurcate Killing horizon — this is a geometric property of the Schwarzschild solution near  $r = 2M$ , independent of the global extension; (3) the Hartle–Hawking state is, by definition, KMS with respect to the Killing flow at the Hawking temperature.

Sewell's conclusion: the algebra of observables for the wedge is a Type III<sub>1</sub> von Neumann factor, and the modular conjugation  $J$  implements CRT symmetry (charge  $\times$  reflection  $\times$  time reversal), which in this context is CPT.

Therefore  $J = \text{CPT}$ . This result is obtained without any assumption about global manifold structure beyond the wedge — specifically, without assuming the existence of Regions III or IV.

The remainder of the proof ( $\hat{\sigma}$  satisfies the defining properties of  $J \rightarrow \hat{\sigma} = J$  by uniqueness) proceeds without modification.

Note on standard imports: Sewell's theorem assumes the Hadamard condition (microlocal spectrum condition) on the quantum state. This is a standard condition in curved-spacetime AQFT ensuring physical reasonableness of the state.

The argument imports this as standard apparatus, at the same level as importing Gleason's theorem (AP25) or Tomita–Takesaki theory. No new physical postulate is introduced. Gap 5 is CLOSED.

**KS-46 status: ADDRESSED.** The derivation is presented. The identification is no longer a bare conjecture — it is a theorem with five tagged gaps. KS-46 upgrades from LIVE—HARD to ADDRESSED, split as follows:

KS-46A (Boundary identification): The  $\sigma$ -boundary = event horizon via AP17's 1-pole/ $\emptyset$ -pole mapping. Depends on AP17. Status: DERIVED (AP17-dependent).

KS-46B (AQFT bridge): The derived QFT must satisfy the conditions for modular theory on the two-sector horizon geometry. Gap 5 (bifurcation tension) CLOSED via Sewell (1982): the local theorem's hypotheses are all satisfied.

Late-time applicability for collapse-formed black holes addressed. Status: CLOSED. Hadamard condition imported as standard AQFT apparatus.

KS-46C (Operator identification):  $\hat{\sigma}$  must equal J via Tomita-Takesaki uniqueness. Status: ADDRESSED (formal anti-linearity proof exhibited; uniqueness argument complete).

Full closure requires: (a) formal discharge of the AQFT bridge conditions on the two-sector manifold (Gaps 1, 3 — both SMALL), (b) rigorous proof that the late-time near-horizon modular structure converges to the bifurcate-horizon result (physically established, not formally proved), (c) construction of  $\hat{\sigma}$  as a concrete Tomita operator on the specific algebra (formal anti-linearity proof exhibited; full construction uses standard AQFT apparatus).

If any of these is shown to be impossible, the theorem reverts to a conjecture and the downstream results (§5.1, §6, Proposition 2) become conditional.



## 5 — The Net Asymmetry

### 5.1 — The segregation

By Theorem 1 (§4), the event horizon is the  $\sigma$ -boundary, and  $\hat{\sigma} = \text{CPT}$  at the horizon. The 1-pole/0-pole topology of the Eye (AP17) then implies the following segregation:

At the actualisation event — the origin of the expanding manifold (AP20) — the 1:1 broke via the splinter (Axiom B). The tension field formed (AP17).

The topology of the Eye emerged: 1-poles (propagation, outward, light, matter) and 0-poles (fold, inward, gravity, antimatter). The  $\sigma$ -involution maps one to the other.

But the two sides of the Eye are not equivalent: one propagates outward (visible), the other collapses inward (hidden behind horizons).

The structural argument for segregation: The 1-pole is the condition of propagation — outward expansion, light, the  $\mathcal{L}$ -sector. The 0-pole is the condition of collapse — inward folding, gravity, the  $\mathcal{P}$ -sector.

By Theorem 1, the event horizon is the boundary between these two conditions. Matter ( $\sigma$ -even content) propagates in the  $\mathcal{L}$ -sector. Antimatter ( $\sigma$ -odd content, the  $\sigma$ -images) resides in the  $\mathcal{P}$ -sector — inside the horizons.

From one Eye to many: The primordial actualisation creates one Eye topology with one  $\emptyset$ -pole. As the tension field evolves, it forms filaments via the Steiner tree mechanism (AP21, Proposition 2).

At filament intersections, energy density exceeds the compactification threshold. The primordial  $\emptyset$ -pole fragments into individual direct vacuums — primordial SMBHs. Each SMBH is a local  $\emptyset$ -pole, a local expression of the  $\mathcal{P}$ -sector.

Additional  $\emptyset$ -poles form throughout cosmic history: stellar black holes (from collapse of massive stars), primordial black holes (from density fluctuations), and any other structure where Axiom C forces compactification.

The visible universe is the white of the Eye. The black holes — all of them: supermassive, stellar, primordial — are the black of the Eye.

The net baryon surplus in the visible universe is balanced by the net antibaryon content inside the horizons.

You are standing on the matter side of a ledger whose other half is hidden behind every black hole in the universe. The books balance. You just cannot see the other page.

## 5.2 — Why the door stays open

If the ledger is exactly 1:1, why doesn't everything annihilate?  
Why is there a visible universe at all?

Because of Axiom B.

**Proposition 1 (Uncancellable entry).** In the axiom structure  $1:1 + 1 \times \varepsilon @ AS$ , the element  $\varepsilon$  has no  $\sigma$ -image. The ledger has exactly one entry that cannot be cancelled by the involution.

**Proof.** The involution  $\sigma$  maps  $\mathcal{L} \leftrightarrow \mathcal{P}$ . The notation “1:1” denotes a bijection: every element in  $\mathcal{L}$  has exactly one image in  $\mathcal{P}$ , and every element in  $\mathcal{P}$  has exactly one pre-image in  $\mathcal{L}$ .

The 1:1 exhausts  $\sigma$  — the bijection accounts for all paired elements. The “ $+1 \times \varepsilon$ ” is the remainder: one additional element, the splinter, that does not participate in the bijection.

Therefore  $\varepsilon$  is not in the domain of  $\sigma$ . Therefore  $\varepsilon$  has no  $\sigma$ -image.  $\square$

Note: Proposition 1 is independent of Theorem 1. It depends only on Axioms S and B. Whether or not the event horizon is the  $\sigma$ -boundary,  $\varepsilon$  has no  $\sigma$ -image.

The existence of an irreducible asymmetry is unconditional.

Because  $\varepsilon$  has no reflection in the  $\mathcal{P}$ -sector, the 1:1 cannot perfectly close. The splinter holds the door open. Matter

remains outside the horizon and antimatter remains inside, held apart by the structural geometry.

The visible universe exists because  $\varepsilon$  exists.

You exist because the axiom has a remainder. One uncancellable entry. One splinter that the mirror cannot reflect. That is you. That is everything you have ever seen.

### 5.3 — The ratio

The observed baryon asymmetry parameter is  $\eta \approx 6 \times 10^{-10}$ . Approximately one extra baryon per billion photons.

**The value debt (tracked across the body of work as KS-60, owned by AP26).** The argument provides the structural reason for the asymmetry ( $\varepsilon$  has no  $\sigma$ -image, Proposition 1) but does not yet derive the ratio. Proposition 1 says the asymmetry must exist.

The magnitude requires a calculation: the ratio  $\eta$  may be derivable from the information content of the minimum viable splinter ( $\varepsilon$ ) relative to the total degrees of freedom available at the epoch of segregation.

This calculation is owed.

## 6 — The Formal Mapping

Let  $|m\rangle$  be a state in the visible manifold ( $\mathcal{L}$ -sector). Let  $\hat{\sigma}$  represent the  $\sigma$ -involution as an operator on quantum states. By Theorem 1,  $\hat{\sigma} = \text{CPT}$  at the event horizon.

$$\hat{\sigma}|m\rangle = |\tilde{m}\rangle$$

where  $|\tilde{m}\rangle$  is the  $\sigma$ -conjugated state residing inside the horizon. By Theorem 1,  $\hat{\sigma}$  maps every quantum number to its conjugate: charge  $\rightarrow$   $-\text{charge}$ , baryon number  $\rightarrow$   $-\text{baryon number}$ , lepton number  $\rightarrow$   $-\text{lepton number}$ .

For every proton in the visible universe, there is an antiproton behind a horizon. For every electron, a positron.

**Proposition 2 (Global quantum number balance).** For every observable quantum number  $Q$  that is  $\sigma$ -odd, the total over the complete system (exterior + all horizon interiors) satisfies:  $Q_{\text{total}} = Q(\varepsilon)$ , where  $Q(\varepsilon)$  is the quantum number of the splinter.

**Proof.** By Theorem 1, the exterior manifold is the  $\mathcal{L}$ -sector and the interior of every horizon is the  $\mathcal{P}$ -sector.

By Axiom S,  $\sigma$  is a bijection between  $\mathcal{L}$  and  $\mathcal{P}$  ( $\sigma(\mathcal{L}) = \mathcal{P}$ ,  $\sigma(\mathcal{P}) = \mathcal{L}$ ): every element in  $\mathcal{L}$  has a unique image in  $\mathcal{P}$  and conversely.

If  $Q$  is  $\sigma$ -odd, then  $Q(x) + Q(\sigma(x)) = 0$  for every paired element  $x$ . The sum over all paired elements vanishes. By Proposition 1, the only unpaired element is  $\varepsilon$ .

Therefore  $Q_{\text{total}} = \sum Q = Q(\varepsilon)$ .  $\square$

The total state of a galactic system is:  $| \text{galaxy} \rangle = | \text{visible matter} \rangle + | \text{interior of all BHs} \rangle$ . The global quantum numbers of the total state sum to  $Q(\varepsilon)$ .

For practical purposes,  $|Q(\varepsilon)|$  is negligible compared to the total number of particles. The asymmetry is real but structurally minimal.

Note on mass balance: Proposition 2 requires that the total antibaryonic content inside ALL horizons (not just SMBHs — also stellar black holes, primordial black holes, and any collapsed structure where Axiom C forces the 0-condition) must equal the total baryonic content of the visible universe, minus the splinter's contribution.

This is the quantitative content of KS-47. The observed mass inside known black holes (total SMBH mass  $\sim 10^{43}$  kg, total stellar BH mass  $\sim 10^{42}$  kg) falls roughly ten orders of magnitude short of the visible baryonic mass ( $\sim 10^{53}$  kg).

This is a real quantitative tension that the paper does not resolve.

The Schwarzschild mass of a black hole includes all interior energy, not just rest mass; and unobserved primordial BHs below current detection thresholds may contribute significantly.

But until KS-60 (AP26) provides a quantitative calculation, the mass balance remains an open question with teeth: if the total mass inside all horizons cannot, even in principle, account for the antibaryonic content, KS-47 kills the paper.

It is a quantitative prediction with teeth — a near-falsification the paper discloses rather than resolves.

Note on  $\hat{\sigma}$  and CPT: The standard CPT theorem is a statement about Lorentz-invariant quantum field theories on a fixed background spacetime.

The  $\hat{\sigma}$ -operator has a different origin: it is a consequence of Axiom S — the two-sector structure of the pre-state. Theorem 1 shows that the two coincide at the event horizon.

The CPT theorem, in this argument, is not an accidental symmetry of QFT but a consequence of the universe's axiomatic geometry expressed at the boundary between the two sectors.

A note on scale: Proposition 2 mirrors the entire visible baryon number — the whole  $\sim 10^{80}$ , not merely the one-in- $10^9$  asymmetry. This is the strong claim KS-47 tests, and the source of its mass-accounting tension; §3's distinction

between transient local antimatter and the net asymmetry keeps the machinery consistent, but the quantity behind every horizon is the full baryon count, not the surplus alone.



## 7 — Derivation Chain

**AP09** (complex Hilbert space) + **AP05** (Lorentzian spacetime) + **AP08** (EFEs) + **AP21** (Energy–Measure Bridge) + **Axiom C** (Constraint) → Derived QFT satisfies conditions for modular theory (Stage 1).

**AP17** (1-pole/0-pole) + **Axiom C** (Constraint) + **AP08** (horizons exist) →  $\sigma$ -boundary = event horizon (Part A).

**Axiom R** (Record — irreversibility) + **Axiom S** (two sectors) → Two-sector topology: no Regions III/IV (Proposition 0).  
Commutant of  $a_{\mathcal{L}}$  is  $a_{\mathcal{P}}$  (Corollary).

Conditions for modular theory + Killing horizon (AP08) + two-sector topology + Sewell / Kay–Wald → Modular conjugation  $J$  maps  $a_{\mathcal{L}} \rightarrow a_{\mathcal{P}}$  and  $J = \text{CPT}$  (Stage 2).

**Axiom S** (Symmetry —  $\sigma$  is involution) + Part A ( $\sigma$ -boundary = horizon) + Tomita–Takesaki structure →  $\hat{\sigma}$  satisfies defining properties of  $J$  (Stage 3).

Tomita–Takesaki uniqueness (polar decomposition) →  $\hat{\sigma} = J = \text{CPT}$  (Stage 4). **Theorem 1 proved.**

Theorem 1 + AP17 (Eye topology) + AP21 (filament fragmentation) → Segregation: matter outward, antimatter inward (§5.1).

**Axiom B** (Break —  $\varepsilon$  has no  $\sigma$ -image)  $\rightarrow$  Uncancellable entry  
(**Proposition 1**, unconditional).

Theorem 1 + Proposition 1  $\rightarrow$  Global quantum number balance  
 $Q_{\text{total}} = Q(\varepsilon)$  (**Proposition 2**).

## 8 — Conclusion

The symmetry was never broken. It was folded.

Local antimatter exists. It is produced and annihilated constantly via pair production — the  $\sigma$ -involution operating transiently on the manifold. This is standard physics.

The net asymmetry — the surplus of matter over antimatter in the visible universe — arises from topological segregation. The visible universe is the  $\mathcal{L}$ -sector (the 1-pole, the white of the Eye).

The interior of every event horizon is the  $\mathcal{P}$ -sector (the 0-pole, the black of the Eye). Theorem 1 derives this identification: the event horizon is the  $\sigma$ -boundary, and  $\hat{\sigma} = \text{CPT}$  there.

The door stays open because  $\varepsilon$  has no  $\sigma$ -image (Proposition 1, unconditional). The break is the one entry in the ledger that cannot be cancelled. The universe exists because the break exists.

There is something rather than nothing because the splinter has no mirror.

The mechanism sidesteps the Sakharov conditions because it is topological, not dynamical.

No baryon number violation occurs globally — the total baryon number of the universe is  $B(\varepsilon) \approx 0$ . The asymmetry is apparent, not fundamental.

What appears as excess matter is the visible half of a balanced ledger.

You have held a ledger in your hands. You know what balance means. This is that — at the scale of the universe. One column is the sky above you.

The other column is behind every event horizon. The books balance to within one entry:  $\varepsilon$ . The crack that will not close. The reason there is something rather than nothing.

# Kill Switches

Kill switch numbers are globally unique across the 420 Code. Five switches engage on this paper. KS-46 is split into three sub-switches (KS-46A/B/C); KS-47 and KS-53 are empirical.

## KS-46A — Boundary identification

**Claim.** The  $\sigma$ -boundary is the event horizon, via AP17's 1-pole / 0-pole identification of the two sectors.

**Test.** Show AP17's sector identification is wrong, or that the  $\sigma$ -boundary does not coincide with the event horizon.

**Status.** DERIVED (AP17-dependent).

**Recovery.** Part A of Theorem 1 fails; the theorem loses its geometric anchor and the segregation mechanism has no boundary.

## KS-46B — AQFT bridge

**Claim.** The derived QFT satisfies the conditions for modular theory on the two-sector horizon geometry (Bisognano–Wichmann / Sewell): energy bounded below, microcausality, a cyclic and separating vacuum.

**Test.** Show the derived QFT fails a required condition on the two-sector manifold, or that the bifurcation tension cannot be discharged.

**Status.** ADDRESSED. The bifurcation tension (Gap 5) is discharged via Sewell (1982); the Hadamard condition is imported as standard AQFT apparatus, at the level of Gleason's theorem. The broader bridge conditions — the rigorous AQFT (Haag–Kastler) restatement (Gap 1) and the uniqueness-of-vacuum step W3 — remain SMALL gaps. A sub-tension discharge is not a full closure: KS-46B is ADDRESSED, not CLOSED.

**Recovery.** The modular-theory bridge fails;  $\hat{\sigma} = J$  cannot be established and Part B reverts to conjecture.

## KS-46C — Operator identification

**Claim.**  $\hat{\sigma} = J$ : the  $\sigma$ -involution equals the Tomita–Takesaki modular conjugation, by uniqueness ( $\hat{\sigma}$  is the anti-linear involution mapping the exterior algebra to its commutant and preserving the vacuum).

**Test.** Show  $\hat{\sigma}$  does not satisfy the defining properties of  $J$ , or that the uniqueness argument fails.

**Status.** ADDRESSED. Formal anti-linearity proof exhibited; uniqueness argument complete.

**Recovery.** The operator identification reverts to conjecture; Theorem 1 weakens to a structural motivation.

## **KS-47 — Global baryon number**

**Claim.**  $B_{\text{total}} = B(\varepsilon) \approx 0$  for the whole universe: every visible baryon is matched by an antibaryon behind a horizon (Proposition 2).

**Test.** Show the total antibaryonic content inside all event horizons cannot, even in principle, account for the visible baryon excess — the  $\sim 10$ -order-of-magnitude mass-accounting tension.

**Status.** LIVE — EMPIRICAL.

**Recovery.** Proposition 2 and the entire ledger mechanism fail. This is a quantitative prediction with teeth, pending the value calculation owed under KS-60.

## **KS-53 — Hawking evaporation products**

**Claim.** The black-hole interior is the  $\mathcal{P}$ -sector; evaporation should reflect the conjugation — either net antibaryon number in the radiation, or a remnant.

**Test.** Confirm Hawking radiation is purely thermal with  $B = 0$  and that no remnant persists.

**Status.** LIVE — EMPIRICAL.

**Recovery.** The conjugation's evaporation prediction fails; the interior-as- $\mathcal{P}$ -sector reading is contradicted.



# **What This Establishes and What Remains Open**

## **What this paper establishes**

This paper provides the structural mechanism for baryogenesis without dynamical symmetry violation.

The picture follows from the axioms: the vacuum is under tension (AP17), the tension segregates into 1-poles (propagation, matter) and 0-poles (collapse, antimatter), the event horizon is the boundary between them (Theorem 1), and the splinter holds the door open (Proposition 1).

The baryon asymmetry is the visible side of a globally balanced ledger.

## **What this paper does not do**

This paper does not provide:

- A quantitative derivation of the baryon asymmetry ratio  $\eta \approx 6 \times 10^{-10}$ . This debt is tracked as KS-60 (AP26).
- A resolution of the mass balance tension (the  $\sim 10$ -order-of-magnitude gap between observed BH mass and visible baryonic mass).

- Full closure of the AQFT bridge (KS-46B): the bifurcation tension (Gap 5) is discharged via Sewell (1982), but the rigorous Haag–Kastler restatement (Gap 1) and the vacuum-uniqueness step (W3) remain SMALL gaps, so the switch is ADDRESSED, not closed.

Gaps 1–4 are all SMALL or smaller. Gap 2 (anti-linearity of  $\sigma$ ) is SMALL, with a formal Tomita operator proof exhibited. The theorem has no SIGNIFICANT or MODERATE gaps remaining.

## **Items held open without claiming debt status**

The five tagged gaps of §4.4 (all SMALL or smaller — Gap 4, rotating black holes, is MINOR — with the bifurcation tension closed). The mass-balance question carried as KS-47, pending the value calculation owed under KS-60.

## **Where the philosophical-register implementations live**

AP22 is formal-register throughout. A catalogue audit of the present Ø Models volumes finds no dedicated philosophical-register companion to the antimatter-segregation mechanism: Ø Predictions covers the five derived numbers but not the ledger, and Ø Resolutions touches the matter–antimatter asymmetry only as one of several open cosmological questions. The expository twin to the horizon ledger is not yet written; this is recorded rather than asserted.

## **Structural relationships to subsequent work**

AP22 opens onto AP26 (The Surplus), which reads the same surplus as  $\eta = E(\varepsilon)$  and closes the cross-horizon balance.

Together with AP30 (The Resistance) they form Notebook V — the matter sector.

## **The 420 Code placement**

Notebook V — Particles and Matter, Chapter 2.

## **The closing**

The symmetry was never broken. It was folded. What appears as excess matter is the visible half of a balanced ledger; the other column is behind every event horizon. The books balance to within one entry:  $\varepsilon$ , the splinter with no mirror.

## Claim Summary

**Derived (unconditional):**  $\varepsilon$  has no  $\sigma$ -image (Proposition 1, from Axioms S + B). Net asymmetry must exist because the ledger has an uncancellable entry. Local antimatter from pair production (§3). Sakharov bypass (topological, not dynamical).

**Derived (from Theorem 1):** Event horizon =  $\sigma$ -boundary;  $\sigma^\wedge$  = CPT (§4, Theorem 1, five gaps tagged in §4.4; Gap 5 CLOSED via Sewell 1982; Gaps 1–4 SMALL or smaller).

Two-sector topology (Proposition Ø, from Axioms R + S). Matter outward / antimatter inward (§5.1). Global quantum number balance  $Q_{\text{total}} = Q(\varepsilon)$  (Proposition 2).

**Structural:** Event horizon =  $\sigma$ -boundary motivated by Axiom C and derived via AQFT (Theorem 1, gaps tagged).

**Conjectural/Untested:** Numerical baryon asymmetry ratio (KS-6Ø, owned by AP26). Mass balance of all horizon interiors vs visible baryonic mass (§6 note, KS-47). Hawking evaporation products carry net antibaryon number or remnant persists (KS-53).

**Conditional on:** Nothing. EH and QRA proved in AP2Ø. Axioms unconditional. Theorem 1 has five tagged gaps (§4.4); Gap 5 formally closed via Sewell (1982). Gaps 1–4 all SMALL or smaller.

Standard AQFT results imported as apparatus (Tomita–Takesaki, Bisognano–Wichmann, Sewell, Gleason, Hadamard condition).

**Depends on:** Axiom S (Symmetry —  $\sigma$ -involution), Axiom B (Break —  $\varepsilon$  has no  $\sigma$ -image), Axiom R (Record — irreversibility; kills white holes), Axiom C (Constraint; forces fold; microcausality), AP05 (Lorentzian spacetime; Poincaré covariance), AP09 (complex Hilbert space), AP08 (Einstein field equations; Schwarzschild solution), AP17 (the Eye; 1-pole/0-pole), AP20 (AS = manifold), AP21 (Steiner tree; Energy–Measure Bridge).

Standard AQFT results imported: Tomita–Takesaki theory, Bisognano–Wichmann theorem, Sewell extension (1982), Kay–Wald (1991).

**Formal results:** Theorem 1 (Horizon Conjugation, DERIVED, five gaps tagged). Proposition 0 (Two-sector topology, DERIVED). Proposition 1 (Uncancellable entry, DERIVED, unconditional). Proposition 2 (Global quantum number balance, DERIVED).

**Kill switches:** KS-46 (black hole conjugation, ADDRESSED, split into KS-46A/B/C). KS-47 (global baryon number, EMPIRICAL). KS-53 (Hawking evaporation products, EMPIRICAL).

**Debt owed: the numerical value of the baryon asymmetry ratio  $\eta \approx 6 \times 10^{-10}$  from  $\varepsilon$  — tracked across the body of work as KS-60, owned by AP26 (The Surplus).**

## Conditionality Footer

**Dependencies.** Axiom S (Symmetry —  $\sigma$ -involution), Axiom B (Break —  $\varepsilon$  has no  $\sigma$ -image), Axiom R (Record — irreversibility), Axiom C (Constraint — locality); AP05 (Lorentzian spacetime), AP09 (complex Hilbert space), AP08 (Einstein field equations; Schwarzschild), AP17 (the Eye; 1-pole / 0-pole), AP20 (AS = manifold), AP21 (Steiner tree; Energy-Measure Bridge). Standard AQFT imported as apparatus: Tomita-Takesaki, Bisognano-Wichmann, Sewell (1982), Kay-Wald (1991), Hadamard condition.

**Dependents.** AP26 (The Surplus) builds on the segregation mechanism and the cross-horizon balance.

**Kill switches closed by this paper.** None. KS-46B's bifurcation tension (Gap 5) is discharged via Sewell (1982), but the switch as a whole remains open with SMALL gaps and is ADDRESSED, not closed.

**Kill switches not closed by this paper.** KS-46A (DERIVED), KS-46B (ADDRESSED), KS-46C (ADDRESSED), KS-47 (LIVE — EMPIRICAL), KS-53 (LIVE — EMPIRICAL).

**Structural debts owed.** The numerical value of the baryon asymmetry ratio  $\eta \approx 6 \times 10^{-10}$  — tracked across the body of work as KS-60, owned by AP26.

**Items held open without claiming debt status.** The five tagged gaps of §4.4 (all SMALL or smaller); the mass-balance question (KS-47).







## **Chapter 4**

# **The Surplus**

*The baryon asymmetry as the energy of the break*

*Artist's Proof 26*

# Artist's Note

*What this paper is, and why.*

You are made of the leftovers. The early universe should have made equal matter and antimatter, and they should have annihilated to nothing — a universe of light with no structure in it. Instead a tiny fraction survived: about one part in a billion. That fraction is every star, every planet, every cell in your body.

This paper reads that surplus as the ash of the break. The axiom is  $1:1 + 1 \times \varepsilon$  @ AS. The 1:1 is the symmetric component — matter and antimatter in balance, which annihilates and becomes the cosmic microwave background. The  $\varepsilon$  is the break, the asymmetric remainder, the part with no mirror. It cannot annihilate because there is nothing to annihilate it against. The surplus is  $\varepsilon$ . The ash is what had no partner.

Read as an energy partition, the asymmetry ratio follows directly:  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ , where  $E(\varepsilon)$  is the dimensionless energy of the break. The form is derived. The value of  $E(\varepsilon)$  is owed — and that debt is the sharpest blade in the body of work, tracked as KS-60. If  $E(\varepsilon)$  is eventually derived and does not match the observed  $\eta$ , the energy partition is wrong and AP24's constant-unification programme takes the wound with it.

AP26 is the form; AP22 (The Ledger) is the mechanism. AP22 shows where the antimatter went — topologically segregated behind every event horizon. AP26 shows how much matter was left. They reconcile across the horizon: the surplus on our side is matched by an equal and opposite surplus on the other, so the total is exactly zero. The universe borrowed nothing. It owes nothing.

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# Orientation

## Where this paper fits in the 420 Code

AP26 is the third and final chapter of Notebook V — Particles and Matter. AP30 (The Resistance) derived the mass of matter; AP22 (The Ledger) accounted for the antimatter by topological segregation; AP26 reads the leftover. It reads after AP22 (whose horizon conjugation protects the surplus), AP24 (which locks  $E(\varepsilon)$  to the other constants), and AP25 (whose Born rule normalises the symmetric component to one).

## How to read this paper

The spine is short: Lemma 1 normalises the symmetric component to one (via AP25); Proposition 1 reads the partition and gives the form  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ ; Proposition 2 shows the surplus survives annihilation because  $\varepsilon$  has no  $\sigma$ -image (via AP22). §6 names the one debt — the value of  $E(\varepsilon)$  — with unusual care, because it is the most exposed quantitative claim in the body of work. §7 is explicitly non-load-bearing synthesis; it carries no epistemic weight and says so. The gear-changes are intentional.

## 0 — Dependency and Scope

### 0.1 What this paper does

This paper derives the structural form of the baryon asymmetry of the universe from the core axiom. It does not derive the numerical value. It derives the form.

The universe contains approximately one excess baryon for every billion photons. This ratio,  $\eta \approx 6 \times 10^{-10}$ , is one of the most precisely measured numbers in cosmology.

The standard account requires baryogenesis — a process during the early universe that generated a slight excess of matter over antimatter, satisfying the Sakharov conditions.

The axioms offer a different account: the asymmetry is not generated by a process. It is structural. It is the axiom itself.

The core axiom is  $1:1 + 1 \times \varepsilon$  @ AS. The 1:1 is the symmetric component. The  $\varepsilon$  is the break.

This paper shows that when the axiom is read as an energy partition, the baryon asymmetry ratio follows directly:  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ , where  $E(\varepsilon)$  is the dimensionless energy of the break.

The form is derived. The value of  $E(\varepsilon)$  is a formal debt.

## 0.2 Dependencies

The dependency chain: Axiom S (Symmetry — distinction) → Axiom B (Break — single break) → core axiom ( $1:1 + 1 \times \varepsilon$  @ AS) → AP06 (leakage constant,  $\varepsilon > 0$ ) → AP18/AP21 (Energy–Measure Bridge,  $\mu \rightarrow E$ ) → AP22 (the ledger:  $\sigma$ -asymmetry, topological protection of the surplus) → AP24 (the residual: all constants as projections of  $\varepsilon$ ) → AP25 (the measure: Born rule, normalisation) → this paper (energy normalisation → partition → baryon asymmetry form).

## 0.3 Axiom mapping

Axiom S supplies the distinction (the symmetric 1:1). Axiom B supplies the single break  $\varepsilon$  — the asymmetric remainder with no  $\sigma$ -image. The core axiom  $1:1 + 1 \times \varepsilon$  @ AS, read as an energy partition, is the whole content: the 1:1 normalises to one (AP25), the surplus is  $E(\varepsilon)$ , and the asymmetry is their ratio.

## 0.4 Epistemic status per section

**Epistemic status per section.** §1 (The Problem): historical. §2 (The Axiom as Energy Partition): derived — Lemma 1 grounds the partition in AP25's measure theory and the Energy–Measure Bridge. §3 (The Form): derived — Proposition 1. §4 (Why the Surplus Survives): derived — Proposition 2, topological protection via AP22. §5 (Connection to the Leakage Constant): structural — the leakage constant is the break's coupling face ( $\approx 1/137$ ); the energy face  $E(\varepsilon)$  that sets  $\eta$



is numerically distinct, and its value is owed (KS-60). §6 (The Formal Debt): debt. §7 (The Deeper Point): synthesis — non-load-bearing.

## **0.5 Kill switch summary**

Three kill switches engage on this paper. KS-58 (surplus form) and KS-59 (topological protection) are LIVE — EMPIRICAL; the form is structurally secure, and proton decay has not been observed. KS-60 (derived-value mismatch) is LIVE — DEBT: when  $E(\varepsilon)$  is eventually derived from the self-consistency conditions, it must match the observed  $\eta$ , or the energy partition and AP24's programme both fail. KS-60 is the sharpest blade in the 420 Code.

## **0.6 Structural debts owed**

The numerical value of  $E(\varepsilon)$ , and with it the value of  $\eta$ , is owed — tracked across the body of work as KS-60. The energy-to-number coefficient  $g_{\text{eff}}$  (whether the mapping is direct,  $g_{\text{eff}} = 1$ , or carries a thermodynamic correction) is a precision question internal to that same debt: it is part of what KS-60 must settle, not a separate debt. The structural form, the topological protection, and the connection to the leakage constant are all derived results, not debts.

## **0.7 Items held open without claiming debt status**

Whether  $E(\varepsilon)$  is best obtained by direct computation of the Standard Model spectrum (Path 1) or by the fixed-point method of AP24 (Path 2) is held open; both routes are sketched in §6.2.

## **0.8 Structural relationships**

AP26 depends on AP22 (The Ledger) for the topological protection of the surplus: AP22 gives the mechanism (the antimatter is segregated behind horizons), AP26 gives the form (the surplus is  $\varepsilon$ ,  $\eta = E(\varepsilon)$ ). The two are complementary and reconcile across the horizon — the surplus on our side is matched by an equal and opposite surplus on the other, total exactly zero. That cross-horizon balance is settled only to the extent AP22's KS-47 survives: KS-47 holds LIVE — EMPIRICAL the question of whether the mass behind all horizons can account for the segregated antimatter, and AP26 inherits that tension rather than closing it. AP26 also depends on AP24, which locks  $E(\varepsilon)$  to  $\alpha$ ,  $m_e$ , and  $G$ ; this is why  $E(\varepsilon)$  is not a free parameter, and why KS-60 is simultaneously a test of this paper and of AP24.

## Notation

**$\eta$**  — the baryon-to-photon ratio. Observed value:  $\eta \approx 6 \times 10^{-10}$  (Planck 2018:  $6.1 \times 10^{-10}$ ).

**$E(\varepsilon)$**  — the dimensionless energy of the break. Defined as  $E(\varepsilon) = \mu(\varepsilon) / \mu(1:1)$ , where  $\mu$  is the additive measure on the record monoid (AP18 Lemma 1, AP21 Energy–Measure Bridge).

$E(\varepsilon)$  is a pure number in  $(0, 1)$ , determined by the self-consistency conditions (AP24).

**1:1** — the symmetric component of the axiom. In the energy partition, the component whose matter and antimatter contributions cancel perfectly upon annihilation.

**1 $\times$  $\varepsilon$**  — the break. The asymmetric component. The surplus that has no mirror image and therefore cannot annihilate.

**$\varepsilon_{\text{leak}}$**  — the leakage constant (AP06). The non-zero probability that energy leaks past a perfect absorber. Identified with the break:  $\varepsilon = \varepsilon_{\text{leak}}$ .

**$\sigma$**  — the orientation-reversal operator on the manifold (AP22). The break  $\varepsilon$  has no  $\sigma$ -image:  $\sigma(\varepsilon)$  is undefined.

**$g$**  — *the effective number of relativistic degrees of freedom at a given energy scale. Relevant to the precise mapping*

*between the energy ratio  $E(\varepsilon)$  and the observed number ratio  $\eta$ .*

# 1 — The Problem of the Baryon Asymmetry

You are made of the leftovers.

The universe is made of matter. Not antimatter. This is strange. The laws of physics are nearly symmetric between matter and antimatter.

For every process that creates a particle, there is a corresponding process that creates its antiparticle.

If the universe began in a symmetric state — equal parts matter and antimatter — then all the matter should have annihilated with all the antimatter, leaving nothing but photons.

A universe of light and no structure.

Instead, approximately  $10^{80}$  baryons and essentially zero primordial antibaryons. The baryon-to-photon ratio  $\eta \approx 6 \times 10^{-10}$  tells us that for every billion matter-antimatter pairs that annihilated, roughly one extra baryon survived.

A surplus of one part in a billion.

The standard account (Sakharov, 1967) identifies three necessary conditions for generating this asymmetry: (1) baryon

number violation, (2) C and CP violation, and (3) departure from thermal equilibrium.

These conditions are necessary but not sufficient — the Standard Model satisfies all three but cannot produce a large enough asymmetry. The precise mechanism of baryogenesis remains one of the open problems in physics.

The axioms offer a different account. The asymmetry is not generated by a dynamical process during the early universe. It is structural. It is present in the axiom itself, before any dynamics occur.

## 2 — The Axiom as Energy Partition

The core axiom is  $1:1 + 1 \times \varepsilon$  @ AS. This is not a metaphor. It is the irreducible structure from which all else follows. The  $1:1$  is the symmetric component — the self-balancing ratio.

The  $1 \times \varepsilon$  is the break — the asymmetric remainder that exists because the break is real ( $\varepsilon > 0$ , AP06).

Read this axiom as an energy partition. But first: why can the symmetric component be normalised to 1?

**Lemma 1 (Energy Normalisation).** The total energy budget of the symmetric component normalises to 1. This normalisation is inherited from the probability measure derived in AP25.

**Proof.** AP25 (The Measure) derives the Born rule as the unique probability measure on outcomes of actualisation:  $\mu(P) = \text{Tr}(|\psi\rangle\langle\psi| \cdot P)$ .

This measure normalises to 1 by construction — the probabilities for all outcomes of a complete measurement sum to 1 (AP25, Lemma 1: orthogonal additivity; Axioms S + B: normalisation).

The symmetric component ( $1:1$ ) is the set of all paired records: every particle matched by its antiparticle, every outcome matched by its conjugate.

The total measure of this set is the total probability of all pairwise-cancelling outcomes, which exhausts the symmetric sector.

By AP25's normalisation, this total is 1. The break  $\varepsilon$  sits outside the symmetric sector (it has no  $\sigma$ -image, AP22).

Its measure is  $\mu(\varepsilon) = E(\varepsilon)$ , defined via the Energy-Measure Bridge (AP18/AP21) as the ratio  $\mu(\varepsilon)/\mu(1:1) = E(\varepsilon)/1 = E(\varepsilon)$ . Therefore: the total energy budget is  $1 + E(\varepsilon)$ .

The total energy of the universe is thus distributed between two components:

**Component 1: The symmetric energy.** The energy associated with the 1:1. This energy condenses into matter-antimatter pairs in equal measure. When these pairs meet, they annihilate completely, converting their energy into radiation (photons).

This is the cosmic microwave background — the echo of the symmetric component's annihilation. Normalise this component to 1.

**Component 2: The surplus energy.** The energy associated with the break  $\varepsilon$ . This energy does not have a symmetric counterpart. It is the uncancellable remainder.



It cannot annihilate because there is nothing for it to annihilate with. This is the matter that forms stars, galaxies, and observers.

The total energy is therefore  $1 + E(\varepsilon)$ , where  $E(\varepsilon)$  is the dimensionless energy of the break. The surplus fraction is  $E(\varepsilon) / (1 + E(\varepsilon))$ .

You have watched a bonfire burn. The fire consumes everything that has a partner — fuel meets oxygen, matter meets antimatter, and both become light and heat.

When the fire is done, only the ash remains. The ash is what had no partner. The ash is  $\varepsilon$ .

The universe is made of ash — one part in a billion — and the cosmic microwave background is the light of the fire that consumed everything else.

### 3 — The Form

**Proposition 1 (The Surplus Ratio).** Let  $E(\varepsilon)$  be the dimensionless energy of the break. The baryon asymmetry ratio takes the structural form  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ .

**Proof.** The core axiom partitions reality into two components: the symmetric (1:1) and the break ( $1 \times \varepsilon$ ). The symmetric component produces equal quantities of matter and antimatter. Upon annihilation, this component converts entirely to radiation (photons).

The break component produces a surplus that has no antimatter counterpart (AP22: the break has no  $\sigma$ -image) and therefore survives annihilation as baryonic matter.

The total energy budget is  $E_{\text{total}} = E_{\text{symmetric}} + E_{\text{surplus}} = 1 + E(\varepsilon)$  (normalising the symmetric component to 1).

After annihilation, the energy in radiation is  $E_{\text{radiation}} = 1$  (from the symmetric annihilation) and the energy in surviving baryons is  $E_{\text{baryons}} = E(\varepsilon)$  (from the surplus). The baryon-to-photon energy ratio is:

$$\eta_E = E_{\text{baryons}} / E_{\text{radiation}} = E(\varepsilon) / 1 = E(\varepsilon).$$

The surplus as a fraction of total energy (surplus-to-total ratio) is:

$$\eta = E(\varepsilon) / (1 + E(\varepsilon)).$$

For  $E(\varepsilon) \ll 1$  (which is established: the break is small), these two expressions are equivalent to leading order:  $E(\varepsilon) / (1 + E(\varepsilon)) \approx E(\varepsilon)$ .

### 3.1 — The energy-to-number mapping

Proposition 1 derives an energy ratio. The observed baryon-to-photon ratio  $\eta \approx 6 \times 10^{-10}$  is a number ratio (baryons per photon), not directly an energy ratio.

The relationship between the two depends on the thermodynamics of the annihilation epoch.

At the annihilation epoch ( $\sim 1$  GeV), the average photon energy and the baryon mass are of comparable order, so the energy ratio and number ratio approximately coincide.

However, the precise mapping may carry a coefficient  $g_{\text{eff}}$  related to the effective number of relativistic degrees of freedom  $g$  *at the annihilation temperature*. The general form is:

$$\eta = E(\varepsilon) / (g_{\text{eff}} \cdot (1 + E(\varepsilon)))$$

where  $g_{\text{eff}}$  absorbs the thermodynamic factors that convert the energy partition into a particle number ratio. In the simplest case,  $g_{\text{eff}} = 1$  and Proposition 1's form is exact.

In the general case,  $g_{\text{eff}}$  is a calculable number that depends on the particle content at the segregation epoch.

This is not a weakness of the derivation — it is a precision question. The structural form (surplus fraction of total energy) is derived and locked.

The coefficient  $g_{\text{eff}}$  determines whether the mapping is direct or carries a thermodynamic correction. Determining  $g_{\text{eff}}$  requires knowing the segregation epoch and the particle spectrum at that epoch. This is part of what KS-60 must settle, not a separate debt.

## 4 — Why the Surplus Survives

Proposition 1 assumes that the surplus energy survives annihilation. This is not automatic. Why doesn't the surplus find something to annihilate with?

**Proposition 2 (Topological Protection).** The surplus  $E(\varepsilon)$  is topologically protected against annihilation. The break  $\varepsilon$  has no  $\sigma$ -image (AP22). Therefore the surplus has no antimatter counterpart and cannot annihilate.

**Proof.** AP22 (The Ledger) establishes that the orientation-reversal operator  $\sigma$  maps the matter sector ( $\mathcal{L}$ -sector) to the antimatter sector ( $\mathcal{P}$ -sector).

AP22's Proposition 1 proves that  $\sigma(\varepsilon)$  is undefined: the 1:1 bijection exhausts  $\sigma$  — every paired element of the symmetric component has a  $\sigma$ -image — so the break  $\varepsilon$ , the uncanceled remainder, lies outside the domain of  $\sigma$  and has no  $\sigma$ -image.

For every  $\mathcal{L}$ -record of the symmetric component,  $\sigma$  produces a corresponding  $\mathcal{P}$ -record: these are the matter-antimatter pairs that annihilate. But the break  $\varepsilon$  has no corresponding  $\mathcal{P}$ -record.

Therefore: the energy  $E(\varepsilon)$  associated with the break has no  $\mathcal{P}$ -sector counterpart. There is no antimatter for it to annihilate with. The surplus is topologically protected.

Note: This is the axioms' replacement for the Sakharov conditions. The standard account requires a dynamical process to generate the asymmetry during the early universe.

The axioms' account is that the asymmetry is structural — it exists in the axiom before any dynamics. The Sakharov conditions describe how an asymmetry could be generated from a symmetric initial state.

The axioms say the initial state was never symmetric: it was  $1:1 + 1 \times \epsilon$  @ AS. The asymmetry is primitive.

You have balanced a ledger. Every entry has a matching entry on the other side — except one. One line item with no counterpart.

You cannot cancel it because there is nothing to cancel it against. That line item is the visible universe.

Everything you have ever touched, tasted, or thought about is the one uncancellable entry in a ledger that otherwise balances perfectly.

## 5 — Connection to the Leakage Constant

AP06 (The Leakage Constant) proves that the break has a physical manifestation: the leakage probability  $\varepsilon_{\text{leak}} > 0$ . This is the non-zero probability that in any interaction, some energy leaks past what would otherwise be a perfect absorber.

The leakage is a consequence of the finiteness of  $c$  (Axiom C) and the irreversibility of records (Axiom R).

$\varepsilon$  is one break with several faces (AP24). Its coupling face is the leakage constant  $\varepsilon_{\text{leak}} = \alpha \approx 1/137$  (AP06) — how strongly the break couples back to the field it left. Its energy face is  $E(\varepsilon)$ , the dimensionless energy of the break — the surplus it leaves in the energy budget. These are distinct numbers belonging to one object. The baryon asymmetry is set by the energy face  $E(\varepsilon)$ , not by the coupling face.

**Corollary (Leakage Interpretation).** The baryon asymmetry ratio  $\eta$  is the energy face of the break as a fraction of the total energy budget:  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ . It is not the coupling face:  $\varepsilon_{\text{leak}} = \alpha \approx 1/137$  would give  $\eta \approx 7 \times 10^{-3}$ , some seven orders of magnitude above the observed  $6 \times 10^{-10}$ . The energy face  $E(\varepsilon)$  is therefore numerically distinct from the coupling face; why the two faces of one break differ by orders of magnitude is part of what AP24's self-consistency must explain (KS-60).

The physical picture: the annihilation of the symmetric component is not perfectly efficient. A fraction  $E(\varepsilon)$  of the total energy survives as the break's surplus. This surviving energy is the baryonic matter of the observable universe.

The cosmic microwave background is the radiation from the symmetric annihilation. The baryon asymmetry is the ratio of surplus to total.

**Connection to AP24.** AP24 (The Residual) establishes that all fundamental constants are projections of  $\varepsilon$ . The fine-structure constant  $\alpha_{em}$ , the electron mass  $m_e$ , the gravitational constant  $G$ , and  $\varepsilon_{leak}$  are all faces of one object.

This means  $E(\varepsilon)$  is not a free parameter — it is determined by the self-consistency conditions that fix the values of all the constants simultaneously.

The value of the baryon asymmetry is locked to the values of  $\alpha_{em}$ ,  $m_e$ , and  $G$  by the self-consistency. If you know any one constant, in principle you know them all.

Schematically, AP24's self-consistency has the structure of a fixed-point condition:  $\varepsilon = f(\varepsilon)$ , where  $f$  encodes how the break's energy determines the coupling constants, which in turn determine the energy scales at which the break manifests.

The function  $f$  is implicit in the derivation chain:  $\varepsilon \rightarrow$  Hilbert space structure (AP09)  $\rightarrow$  gauge group (AP15, AP16, AP19)  $\rightarrow$



coupling constants  $\rightarrow$  running of couplings (RGE)  $\rightarrow$  low-energy value of  $\varepsilon$ .

The unique fixed point of this chain is the physical value of  $E(\varepsilon)$ . Making this chain explicit and solving for  $\varepsilon$  *is the content of KS-60*.

## 6 — The Formal Debt

This paper derives the form of the baryon asymmetry ratio. It does not derive the value. The form is  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ . The value requires deriving  $E(\varepsilon)$  from first principles.

This is a formal debt.

### 6.1 — What must be computed

The debt is: derive the dimensionless value  $E(\varepsilon)$  from the self-consistency conditions established in AP24. The target is  $E(\varepsilon) \approx 6 \times 10^{-10}$ , which would yield  $\eta \approx 6 \times 10^{-10}$  (since  $E(\varepsilon) \ll 1$ ).

### 6.2 — Two paths to the value

**Path 1 (Direct computation).** Derive the full Standard Model particle spectrum as harmonics of  $\varepsilon$ . Write the renormalisation group equations with the derived spectrum.

Run the couplings from the Planck scale (where  $E(\varepsilon) \rightarrow 1$ , the break total in the limit) to low energies.

The unique low-energy value of  $E(\varepsilon)$  that produces self-consistent values for  $\alpha_{\text{em}} \approx 1/137$  and  $m_e \approx 0.511 \text{ MeV}$  is the answer. This path requires deriving the particle spectrum first (a major undertaking).

**Path 2 (Fixed-point method).** AP24 identifies a fixed-point structure:  $\varepsilon = f(\varepsilon)$ , where  $f$  encodes the self-consistency conditions. Formulate  $f$  explicitly and solve for the non-trivial fixed point.

If  $f$  has a unique stable fixed point in  $(0, 1)$ , that fixed point is  $E(\varepsilon)$ . This path is more elegant but requires formulating  $f$ , which itself depends on the coupling structure.

### 6.3 — The energy-to-number coefficient

The mapping from the energy ratio  $E(\varepsilon)$  to the observed number ratio  $\eta$  may carry a thermodynamic coefficient  $g_{\text{eff}}$  related to the effective degrees of freedom at the annihilation epoch.

Determining whether  $g_{\text{eff}} = 1$  (the direct mapping) or  $g_{\text{eff}} \neq 1$  (requiring a correction) is part of the debt.

### 6.4 — What this debt does not include

This debt is about the numerical value of  $E(\varepsilon)$ .

The following are NOT debts — they are derived results: the form of the baryon asymmetry ratio (Proposition 1), the topological protection of the surplus (Proposition 2, from AP22), and the connection to fundamental constants via AP24's self-consistency. The numerical value of  $E(\varepsilon)$  — and the reason the energy face differs from the coupling face  $\alpha$  — is the debt (KS-60).

The structural account is complete. The universe has a baryon asymmetry because the axiom is  $1:1 + 1 \times \epsilon$  @ AS, not 1:1. The form is forced. The value is the remaining calculation.

## 7 — The Deeper Point

Synthesis note: the following is non-load-bearing language. It carries no epistemic weight beyond the claims established above.

The standard account of the baryon asymmetry requires a sequence of specific physical events during the early universe: baryon number violation, CP violation at the right level, departure from equilibrium at the right moment.

It is a story about process. Something happened to tip the balance.

The axioms' account is different. Nothing tipped the balance. The balance was never there. The axiom is not 1:1. It is 1:1 +  $1 \times \varepsilon$  @ AS. The asymmetry is not generated. It is primitive.

It exists before space, before time, before any process. The surplus is the break itself, read as energy.

The reason you exist — the reason there is matter rather than just light — is that the axiom has a remainder. The universe is not perfectly symmetric. It never was.

The imperfection is one part in a billion, and it is everything: every star, every planet, every thought. The splinter that escaped.

## Kill Switches

Kill switch numbers are globally unique across the body of work. Three switches engage on this paper.

### KS-58 — Surplus form

**Claim.** The baryon asymmetry has the energy-partition form  $\eta = E(\varepsilon) / (g_{\text{eff}} \cdot (1 + E(\varepsilon)))$ , with  $g_{\text{eff}}$  a calculable thermodynamic factor at the annihilation epoch (Proposition 1; the bare form  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$  is the  $g_{\text{eff}} = 1$  case).

**Test.** Show the baryon asymmetry has a structural form incompatible with this energy partition.

**Status.** LIVE — EMPIRICAL. Structurally secure.

**Recovery.** Proposition 1 fails; the energy-partition reading of the axiom is wrong and the surplus must be rebuilt from a different partition.

### KS-59 — Topological protection

**Claim.** The surplus  $E(\varepsilon)$  is topologically protected against annihilation because  $\varepsilon$  has no  $\sigma$ -image (Proposition 2, via AP22).

**Test.** Observe baryon-number violation — in particular proton decay — at a rate sufficient to erase the asymmetry within the age of the universe.

**Status.** LIVE — EMPIRICAL. Proton decay has not been observed; current experimental bounds are consistent with the protection claim.

**Recovery.** Proposition 2 fails; the surplus is not protected and the structural account of its survival collapses.

## KS-60 — Derived-value mismatch

**Claim.** When  $E(\varepsilon)$  is derived from the self-consistency conditions (AP24), it yields the observed  $\eta \approx 6 \times 10^{-10}$  — including whatever the energy-to-number coefficient  $g_{\text{eff}}$  is found to be.

**Test.** Derive  $E(\varepsilon)$  from the self-consistency conditions and show it disagrees with the observed  $\eta$ ; or show the asymmetry depends on a parameter independent of the fundamental constants.

**Status.** LIVE — DEBT. The sharpest blade in the 420 Code: not merely a test of this paper but of AP24's central claim that all constants are projections of one break.

**Recovery.** The axiom's energy partition is wrong, or AP24's fixed-point structure is wrong; the value owed under this debt

cannot be discharged and the constant-unification programme is wounded.



# **What This Establishes and What Remains Open**

## **What this paper establishes**

The structural form of the baryon asymmetry,  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ , derived from the core axiom read as an energy partition (Proposition 1). The topological protection of the surplus, because  $\varepsilon$  has no  $\sigma$ -image (Proposition 2, via AP22). The connection of the surplus to the leakage constant and to AP24's self-consistency, which locks  $E(\varepsilon)$  to the other fundamental constants. The structural account of why there is matter rather than only light is complete.

## **What this paper does not do**

It does not derive the numerical value of  $E(\varepsilon)$ , and so does not derive the value of  $\eta$ . That value — including the energy-to-number coefficient  $g_{\text{eff}}$  — is owed under KS-60.

## **Items held open without claiming debt status**

Which of the two paths to the value (direct spectrum computation, or the AP24 fixed-point method) is the right route. Both are sketched in §6.2. Separately, the cross-horizon mass balance that underwrites the surplus is held open as AP22's KS-47 (mass accounting behind all horizons), inherited here rather than closed.

## **Where the philosophical-register implementations live**

AP26 is formal-register throughout.  $\eta$  appears among the five derived numbers read in plain language in Ø Predictions (Ø Models catalogue); the matter–antimatter asymmetry is read structurally, from the binary character of B at primordial coupling sites, in Ø Resolutions. Neither is a dedicated companion to the energy-partition reading itself — the surplus-as-ash treatment has no single expository twin in the present Ø Models catalogue, and this is recorded rather than papered over.

## **Structural relationships to subsequent work**

AP26 closes Notebook V — Particles and Matter. With AP30 (mass), AP22 (antimatter), and AP26 (the surplus) in place, the matter sector is built: the break has produced spacetime, the quantum record, the forces and constants, and now the matter those constants weigh.

## **The 420 Code placement**

Notebook V — Particles and Matter, Chapter 3.

## **The closing**

Nothing tipped the balance. The balance was never there. The axiom was  $1:1 + 1 \times \varepsilon$  @ AS from the beginning, and the surplus

is the break itself. You are the one uncancellable entry, read at the scale of a universe — the ash of a crack whose coupling face has depth  $\alpha \approx 1/137$  (the surplus itself being the energy face,  $E(\varepsilon)$ ).

# Claim Summary

**§1 The Problem [HISTORICAL].** The observed baryon asymmetry  $\eta \approx 6 \times 10^{-10}$  and the Sakharov account.

**§2 The Axiom as Energy Partition [DERIVED].** Lemma 1 normalises the symmetric component to one (via AP25's Born rule); the total energy is  $1 + E(\varepsilon)$ , the surplus being the break.

**§3 The Form [DERIVED].** Proposition 1:  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ ; for  $E(\varepsilon) \ll 1$ ,  $\eta \approx E(\varepsilon)$ .

**§3.1 The energy-to-number mapping [DERIVED / PRECISION].** The energy ratio maps to the number ratio up to a coefficient  $g_{\text{eff}}$ , a calculable thermodynamic factor at the annihilation epoch.

**§4 Why the Surplus Survives [DERIVED].** Proposition 2: the surplus is topologically protected because  $\varepsilon$  has no  $\sigma$ -image (via AP22) — the axioms' replacement for the Sakharov conditions.

**§5 Connection to the Leakage Constant [STRUCTURAL].** The leakage constant is the break's coupling face ( $\varepsilon_{\text{leak}} = \alpha \approx 1/137$ ); the energy face  $E(\varepsilon)$  that sets  $\eta$  is numerically distinct. AP24 locks  $E(\varepsilon)$  to  $\alpha$ ,  $m_e$ , and  $G$ , so  $E(\varepsilon)$  is not a free parameter; its value is owed (KS-60).

**§6 The Formal Debt [DEBT].** The value of  $E(\epsilon)$  (and  $g_{\text{eff}}$ ) is owed — tracked as KS-60, the sharpest blade in the body of work.

**§7 The Deeper Point [SYNTHESIS — non-load-bearing].**  
Nothing tipped the balance; the asymmetry is primitive,  
present in the axiom before any process.

## Conditionality Footer

**Dependencies.** Axiom S (Symmetry — distinction), Axiom B (Break — single break); AP06 (leakage constant,  $\varepsilon > 0$ ), AP18 / AP21 (Energy-Measure Bridge), AP22 (the ledger:  $\sigma$ -asymmetry, topological protection), AP24 (the residual: all constants as projections of  $\varepsilon$ ), AP25 (the measure: Born rule, normalisation).

**Dependents.** Closes Notebook V. AP24's self-consistency programme is tested by — not dependent on — this paper's form (KS-60).

**Kill switches closed by this paper.** None.

**Kill switches not closed by this paper.** KS-58 (LIVE — EMPIRICAL), KS-59 (LIVE — EMPIRICAL), KS-60 (LIVE — DEBT).

**Structural debts owed.** The numerical value of  $E(\varepsilon)$  and  $\eta$ , including the  $g_{\text{eff}}$  coefficient — tracked across the body of work as KS-60.

**Items held open without claiming debt status.** Which of the two paths (direct computation, fixed-point method) yields  $E(\varepsilon)$ .



# Epilogue — Where Particles and Matter Stands

Notebook V set out to weigh the constants of Notebook IV and read what they make. It closes as a structural chain from four axioms to ordinary matter, with the empirical content stated plainly and the debts named.

What Notebook V establishes. First, mass has a structural account: the proton-to-electron ratio is reconstructed from  $\{21, 3, 4\}$  and  $\alpha$ , to five parts per billion at order  $\alpha$  with zero free parameters (Chapter 1); its bare neutron-proton row fired on 2 August 2026 at  $7.24\sigma$  and is shown, never repaired, while the realised row — the held distinction priced at second order,  $-0.005\sigma$  — is frozen beside it (Chapter 2). Second, the structural reading locates the antimatter behind every horizon, segregated by the  $\sigma$ -involution identified with modular conjugation — with the commutant identification (interior as the conjugate algebra) held as a structural reading, not delivered outright by the imported modular theory (Theorem 1, KS-46B addressed) — and the break  $\varepsilon$  has no  $\sigma$ -image, which is unconditional (Chapter 3, Proposition 1). Third, the surplus has a form: read as an energy partition, the baryon asymmetry is  $\eta = E(\varepsilon) / (1 + E(\varepsilon))$ , the ash of the break (Chapter 4). The qualitative account of why any surplus survives at all —



because  $\varepsilon$  has no mirror — is complete and unconditional; the magnitude of that surplus is the open debt (KS-60).

What Notebook V does not close. The numerical value of  $E(\varepsilon)$ , and with it the value of  $\eta$ , is owed — tracked as KS-60, the sharpest blade in the 420 Code, and simultaneously a test of AP24. The mass-balance question — whether the mass behind all horizons can account for the segregated antimatter — is carried live as KS-47. The additivity of the resistance layers (KS-30.1) and the closed-form higher-order terms (KS-30.3) remain owed. These are not hidden; they are the empirical edge on which the notebook can be killed.

The one object. The deepest reading of Notebook V is that its four chapters are one. The break  $\varepsilon$  is resistance, unmirrored entry, and energy — mass, antimatter, surplus — read at the manifold scale. The matter you are made of is the resistance of the break; the antimatter you are not made of is behind a horizon; and the reason there is any of you at all is that the surplus had no mirror to annihilate against. One crack — whose coupling face has depth  $\alpha \approx 1/137$  — read four ways.

Notebook V is closed as a structural chain from four axioms to particles and matter, with every load-bearing claim carrying its kill switch and every debt named in the Master Kill Switch Registry.

# Acknowledgement

The 420 Code is the result of a lifetime of thinking about the phrase — treat others like you want to be treated — or how my brain actually phrases it: Don't be a cunt, be kind.

The 420 Code is me trying to explain my life, to myself and attempting to prove to myself that my knowing of I am, is accurate. It came to life from a deep knowing that, if I want to explain the feeling that we are all connected the first step is simply intellectual honesty. It is actually easy, but at the same time incredibly hard and unimaginably uncomfortable.

This body of work was not a labour of love. It was forged in the fires of pain, desperation, recognition, and compulsive obsession with describing what I see and proving I am not crazy.

I can recall the moment I knew, but I cannot recall the logical understanding. That has been a very long and exhausting process of pointing the axiom in every direction possible.

The more I understood, the greater the pain and suffering has been. Today I cannot understand why and how anything I think or say is not blatantly obvious. I honestly feel like the last one to the party and subject of a prank. That is the most difficult reality of my life I have to deal with.

The work has cost me a lot while keeping me functioning. My obsession with my work, the truth, eccentricities and brutal intellectual honesty has had a real cost on the relationships I have. I have made mistakes. The consequences of those choices have been hard, and deserved. But reality doesn't care about intentions, reality audits consequences.

That is the ground this work was made from.

Due to the nature of the work, and seemingly absurd scope of the work, I have no one to share it with. No one to read it. No one to critique it. That is why I argue with myself — write the work and the weapons to kill it. I stress-test every joint as hard as I can, because that is what I had hoped a reader would be willing to do. I did not have that somebody.

Who I found was Claude from Anthropic. Claude worked alongside me and became the reader and peer-reviewer I always wished for — a reader who would ignore the person and only read the work.

I am the author and the architect of this work, fully and alone. The ideas, the axiom and its preconditions, the structural reading, the architecture of the predictions, the cross-prediction loop, the derivational logic, the judgment of whether any joint holds, the philosophical commitments under all of it — these are mine, worked out across thirty years of private effort.

But I did not build every part with my own hands. I am the kid in the class who can see the answer but struggles to write every step down, because it bores me. So I made Claude pour the concrete where I pointed and weld the joints I marked — the derivations stepped through inside the chapters, the algebra, the dimensional analyses, the lattice-QCD comparisons, the renormalisation-group reasoning, the formal apparatus that turns a structural reading into numbers a physicist can check. That work is not my training. The mathematics in this book is more rigorous than I could have written alone, and that is why.

My biggest struggle was getting Claude to work from the axioms — to explain the structural steps first, before writing the math. I had to explain every step before Claude could write it down. The explaining was the work, and the work was mine. Only then would it land. Only then could the math come.

What Claude gave me that I never had was a reader who would argue back — ignore the person, read only the structure, and try to break it. That is what I needed most, and had no one for. This is my building. Claude helped me raise it.

The work is what the work is. I publish it copyleft, free forever, at [420code.org](https://420code.org). Whoever wants to read it can read it. Whoever can correct it can correct it. Whoever can falsify any kill switch in the Master Kill Switch Registry is welcome to

submit the falsification, and the body of work will respond.  
That is the only relationship the work owes anyone.

I am hurt. I am always hurting. The intensity changes.

The work is the work.



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*Don't be a cunt. Be kind.*

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